

PCB Design Techniques

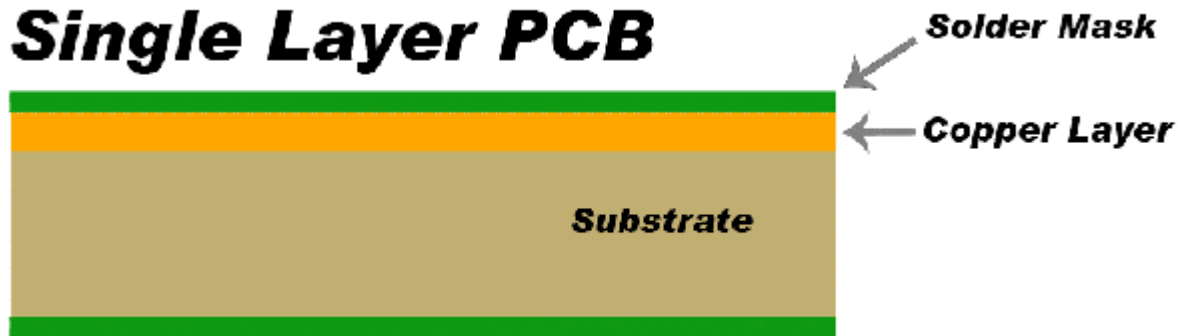
Furkan Karakaya

Outline

- Physical Integrity
- Objects on a PCB
- PCB Manufacturing
- PCB Design Tool Concepts
- Special Design Techniques
- Demo

Layers on a PCB

Single Layer PCB

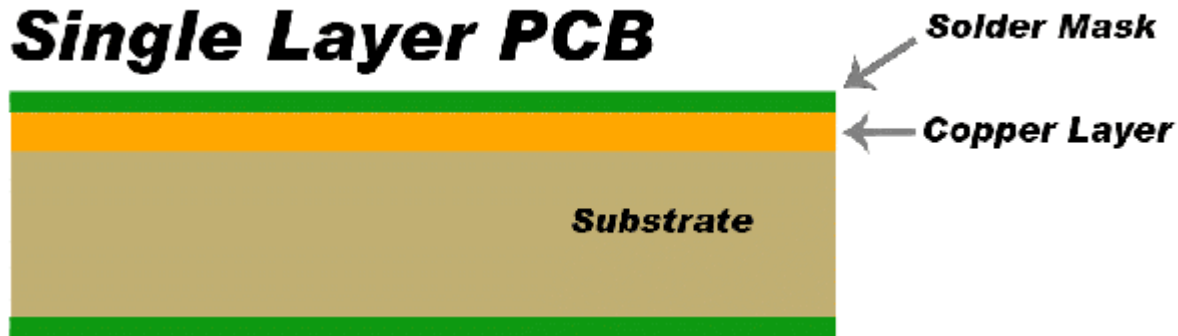


Double Layer PCB



Layers on a PCB

Single Layer PCB



What could be the percent thickness of the copper on a standard PCB?

~2.2%

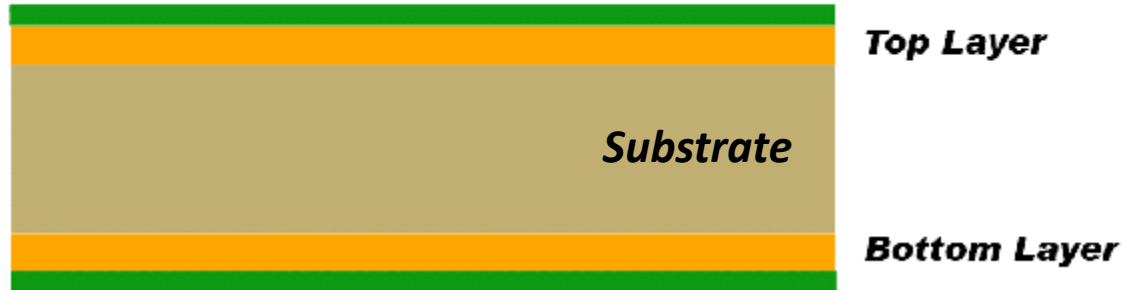
Copper

~97.8%

Substrate

Substrate Material

Double Layer PCB



Important parameters for the substrate:

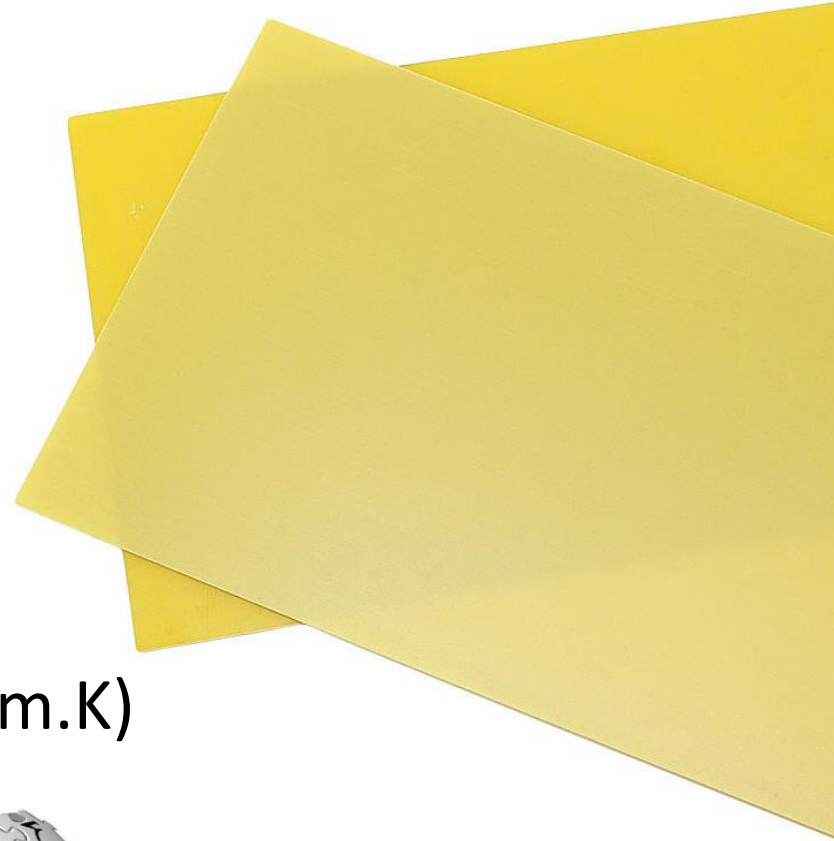
- Bond strength
- Dielectric strength
- Thermal conductivity

Substrate Material

FR-4 is the mostly used substrate

- Bond strength: $> 1000 \text{ kg}$
- Dielectric strength: 20 MV/m
- Thermal conductivity: 0.29 W/(m.K)

WIKIPEDIA
The Free Encyclopedia



Alternative Substrate Materials

$$Capacitance = \frac{k\epsilon_0 A}{d}$$


4.25 – 4.55
for FR-4

High capacitivity might lead to problems for GHz operations.

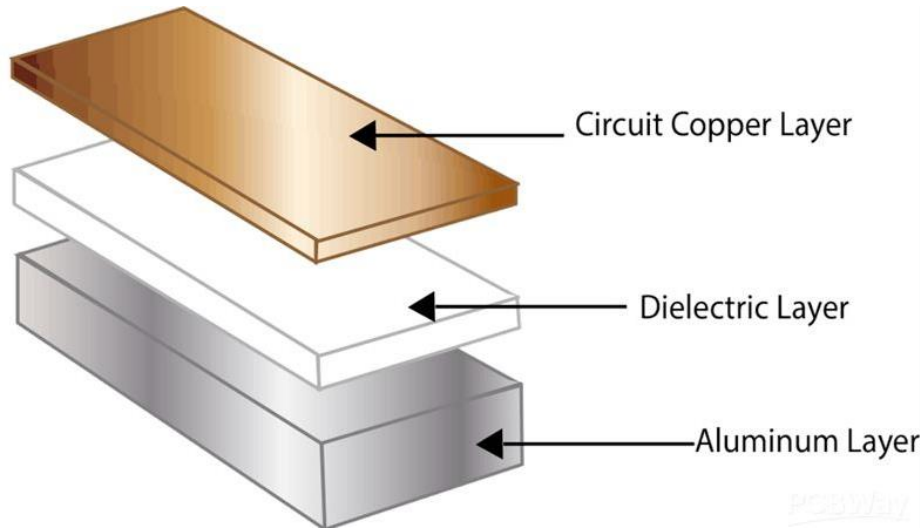
Polytetrafluoroethylene (PTFE) or simply **teflon** is a better choice for high frequency PCBs.

Dielectric constant: **2.5 – 2.8**

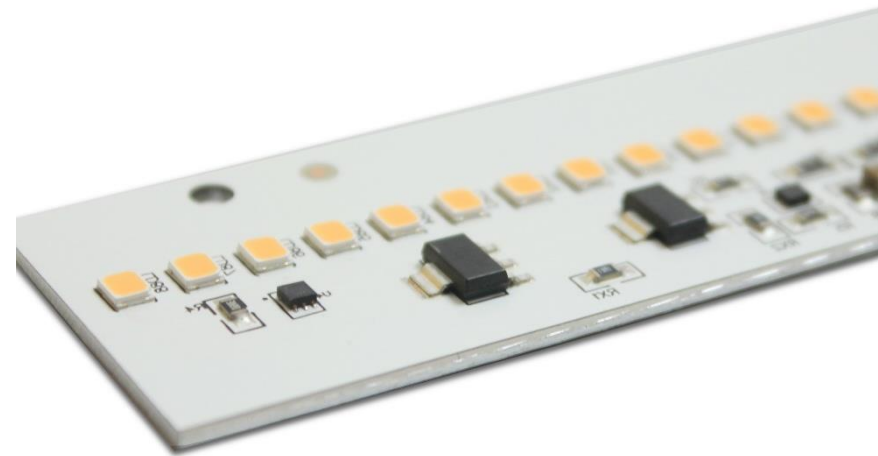
Alternative Substrate Materials

$$\text{Thermal resistivity} = \frac{l}{\sigma A}$$

0.29 W/(m.K)
for FR-4



Metal Core PCB

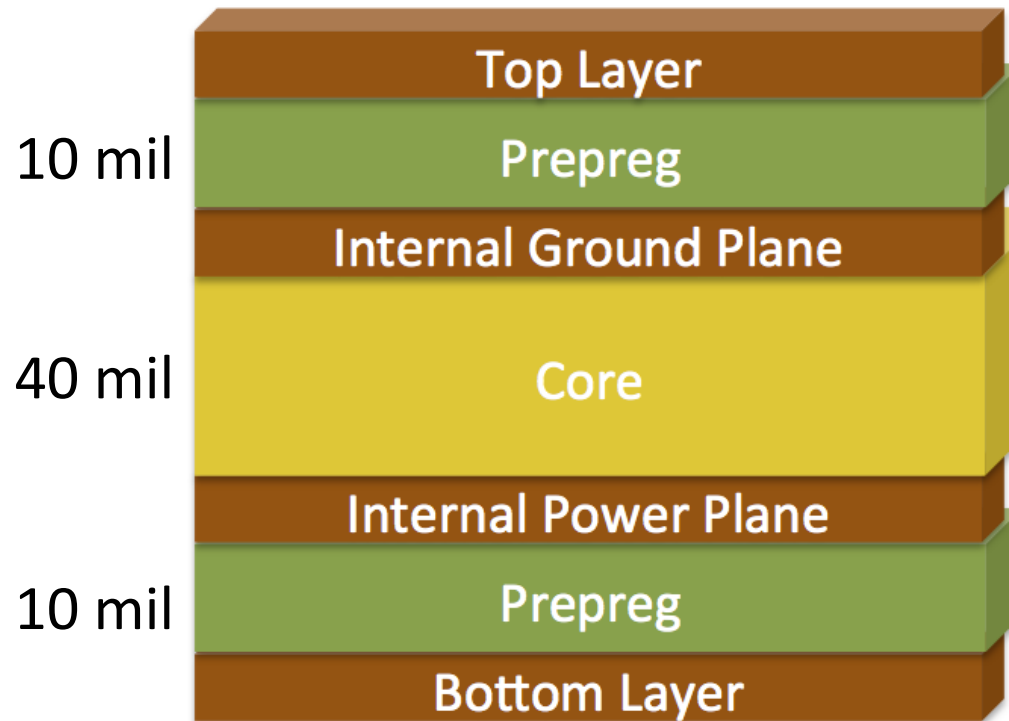


2 W/(m.K) for MCPCB

Conductive Layer

A standard PCB has a copper layer with **1 oz/ft²** thickness or **35 μm**.

Copper thickness
might be increased
up to 13 oz, but...



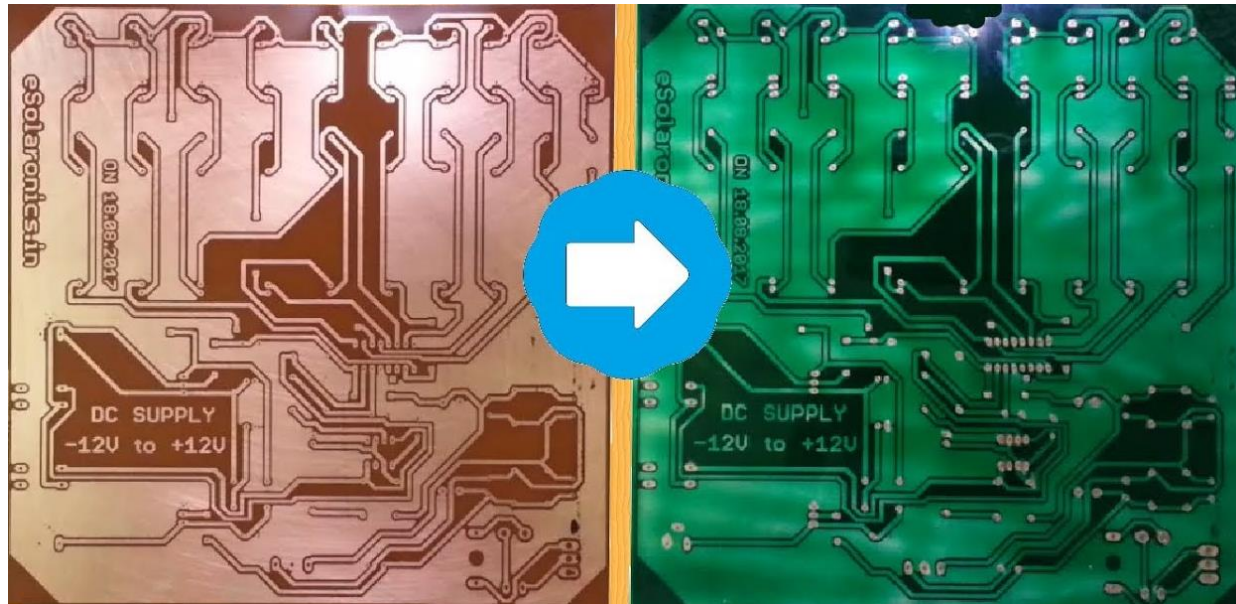
Conductive Layer

Copper thickness might be increased up to 13 oz, but

- The cost increases significantly
- Thicker inner layers increases cost much more
- The minimum distance between side-by-side traces increase

You should be aware of **capabilities of the manufacturer.**

Solder (Stop) Mask



Solder (Stop) Mask



Solder (Stop) Mask

It is effective against

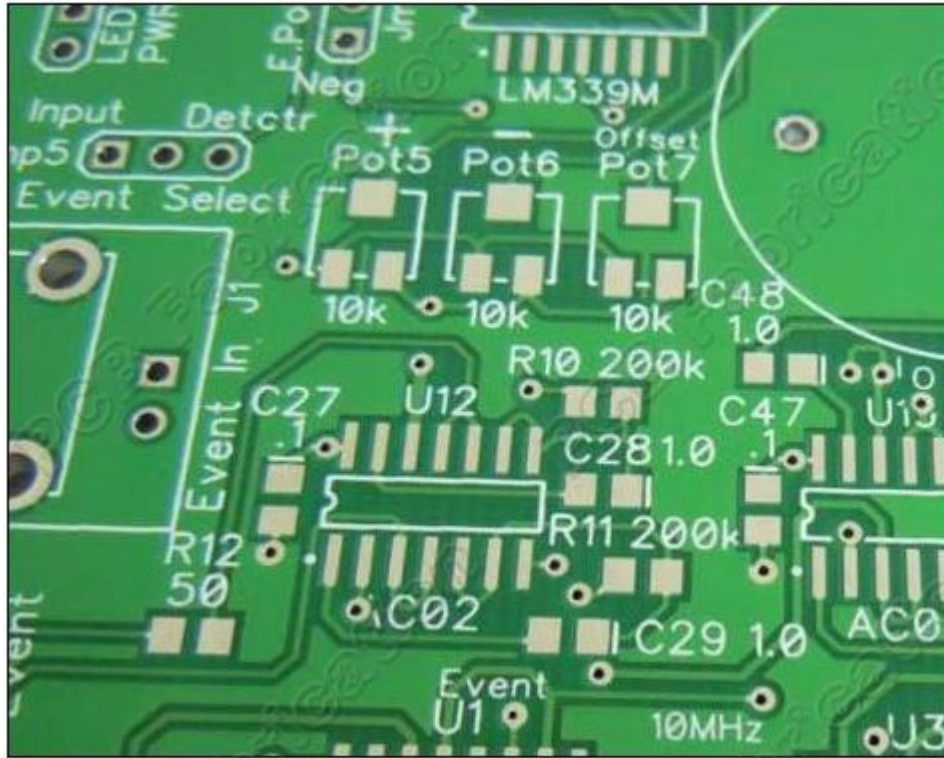
- Corrosion
- Inaccurate soldering

It enables

- Reflow soldering
- Insulation between traces
- Different colors



Silk Screen

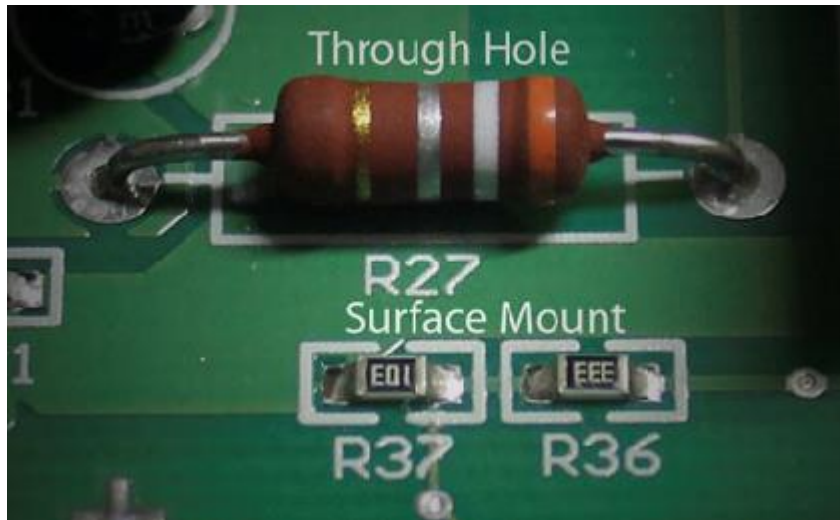


Numbers, logos, instructions are given on the silk screen

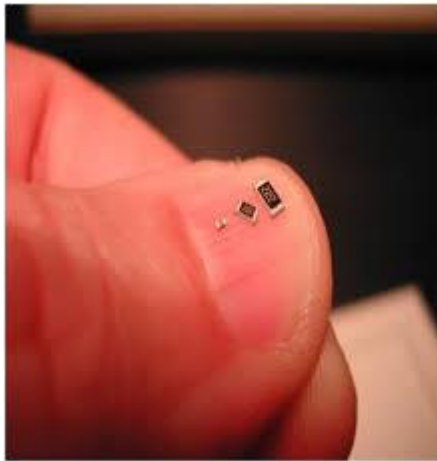
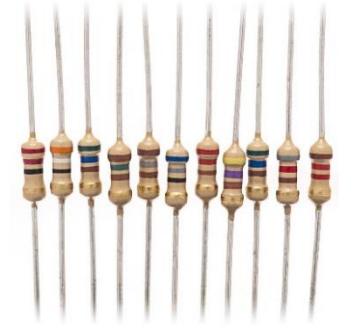
Even your advisor's photo

Objects on the PCB

Pads

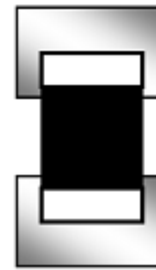
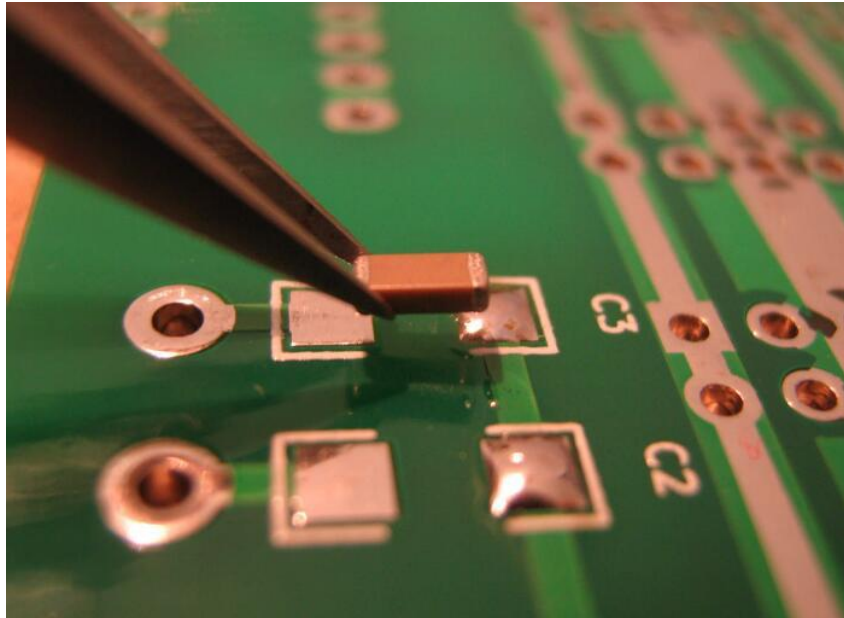


vs.



Pads

How to decide ideal pad size?



Density Level A
Very Robust
Solder Joint



Density Level B
General Purpose
Solder Joint

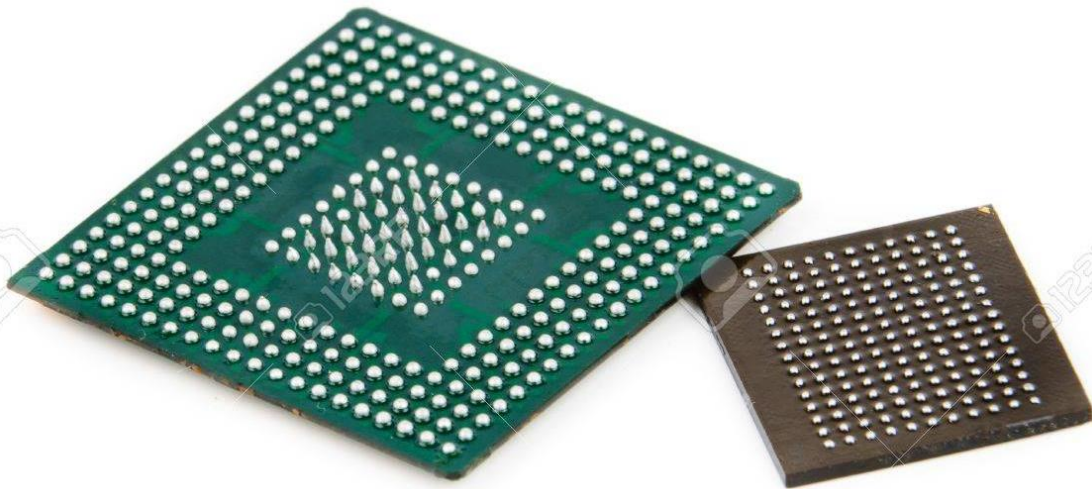


Density Level C
Minimal Solder Joint
High Density Applications

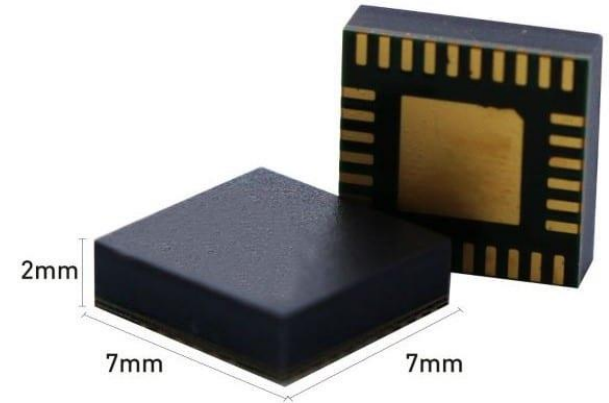
- You should have a plan for soldering
- Check the datasheet of the component for the pad size

Pads

Some of the components might not be soldered by hand!

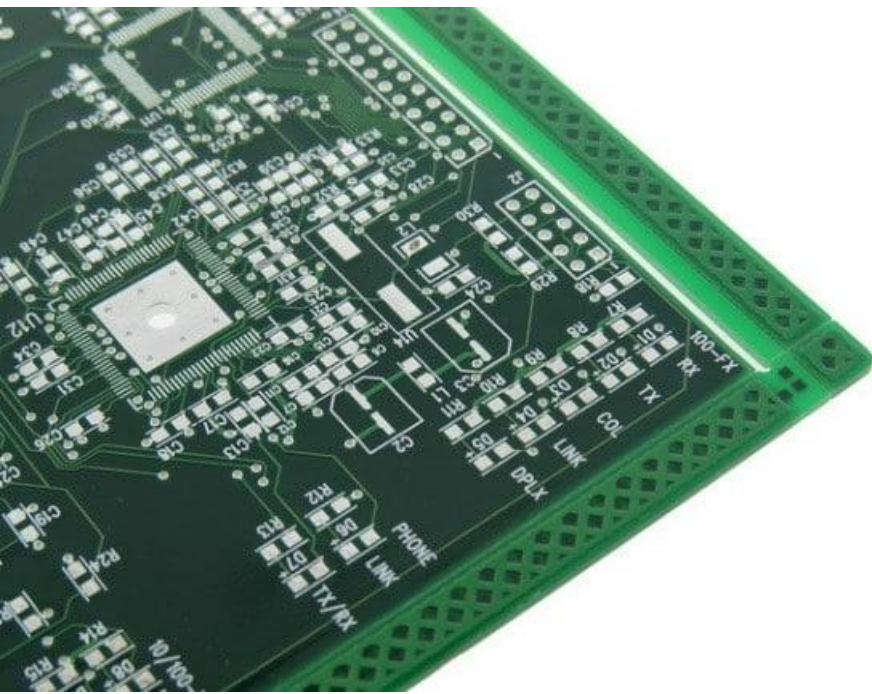


BGA Package

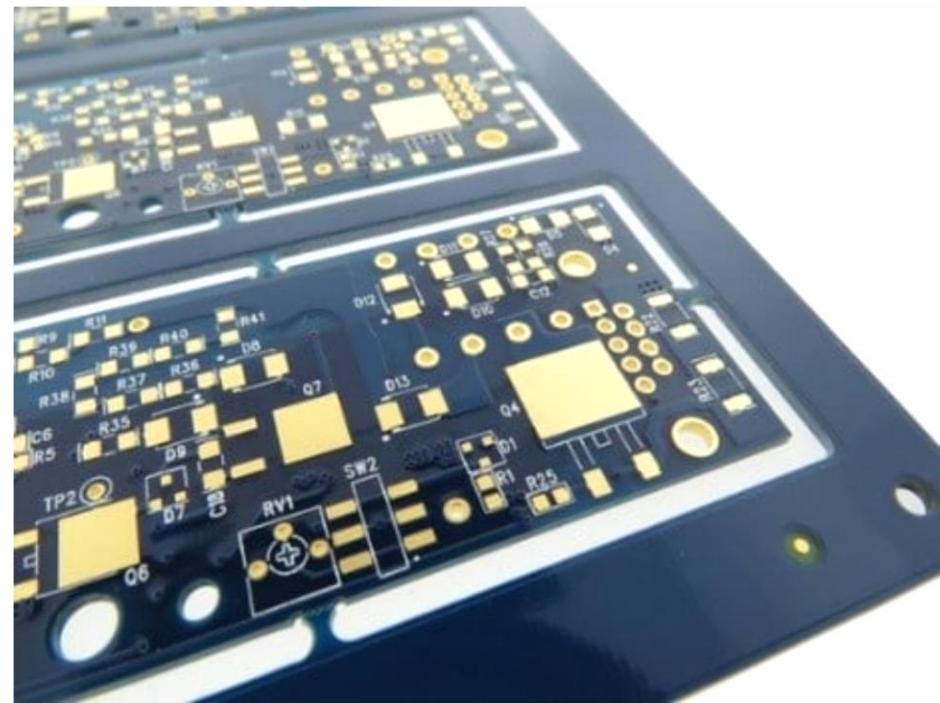


LGA Package

Pads – Surface Finishing



HASL (Hot air solder leveling)

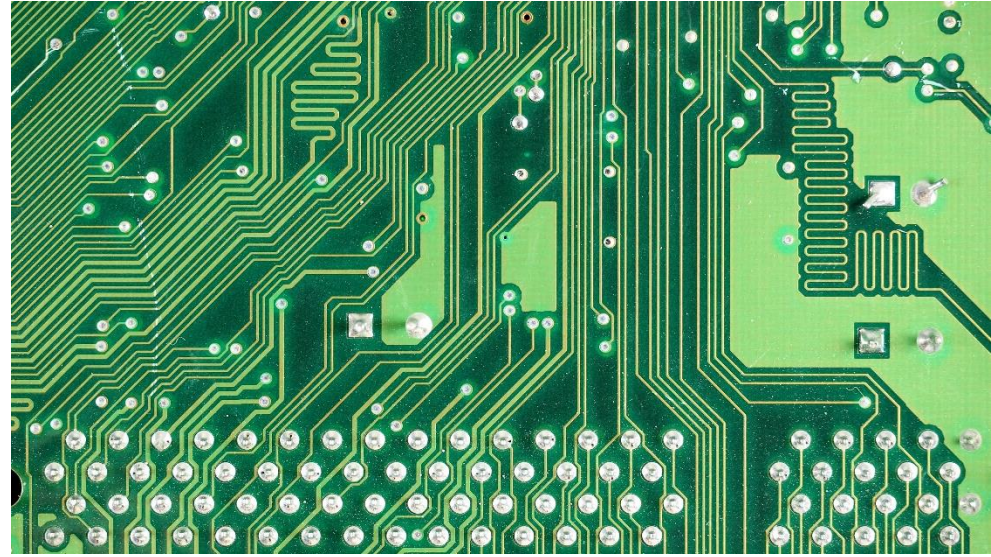


ENIG (Electroless Nickel Immersion Gold)

The surface quality might affect components performance.

Traces

Thin lines of copper



Each trace has resistive and inductive character ★

Traces

How thick it should be?

$$R = \frac{\rho l}{A} \text{ where } \rho = \rho_0(1 + \alpha(T - T_0))$$

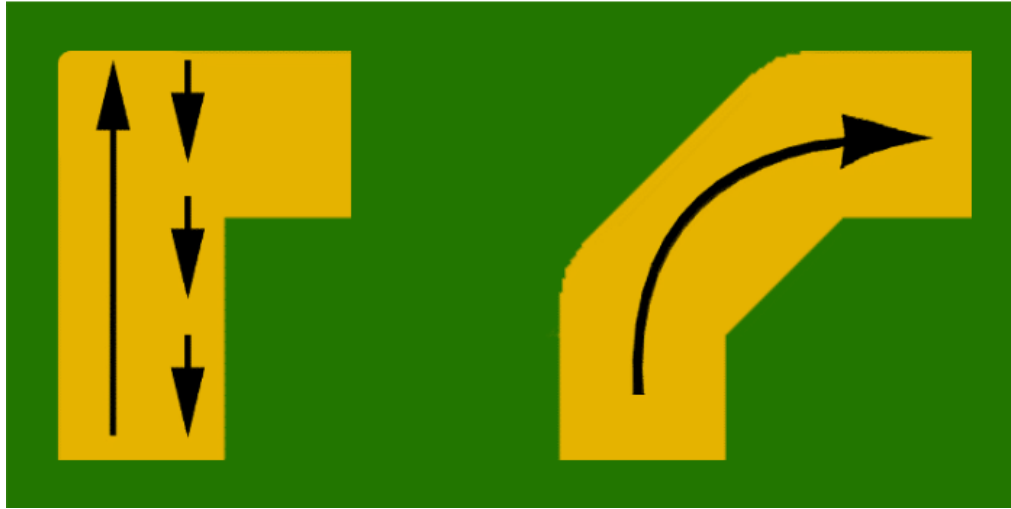
or simply

[Online Resistance Calculator](#)



Traces

Why do we mostly use 45° corners?

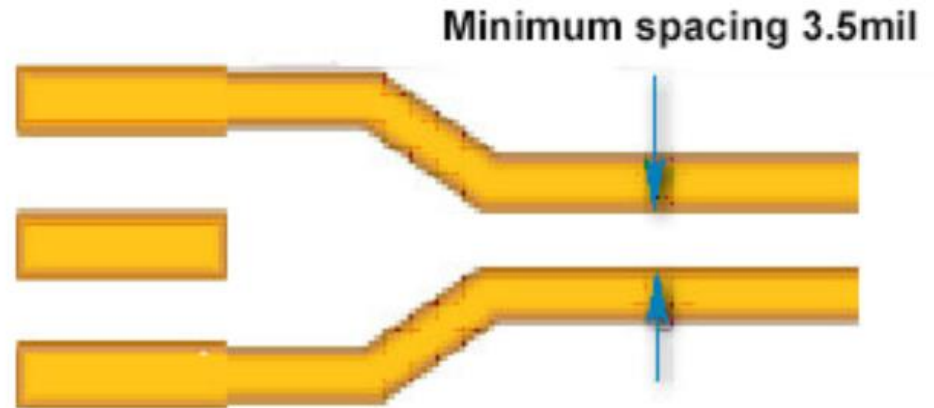


- Ultra high-frequency signals can get reflected off the back of the angle which results in EMI
- 45° corners can be engraved much more accurately

Traces

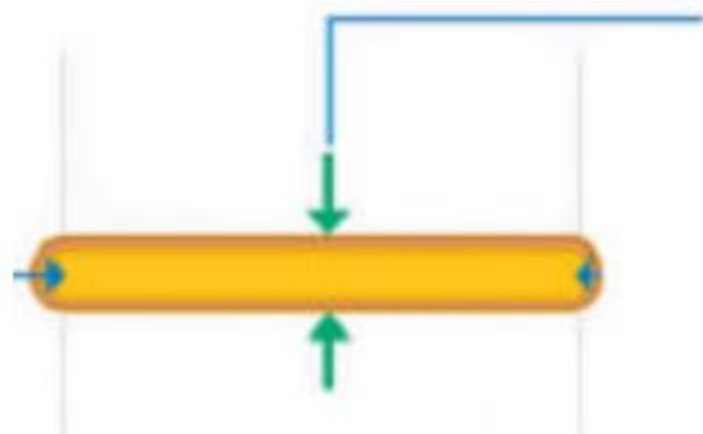
Check the capabilities of the manufacturer

Minimum spacing



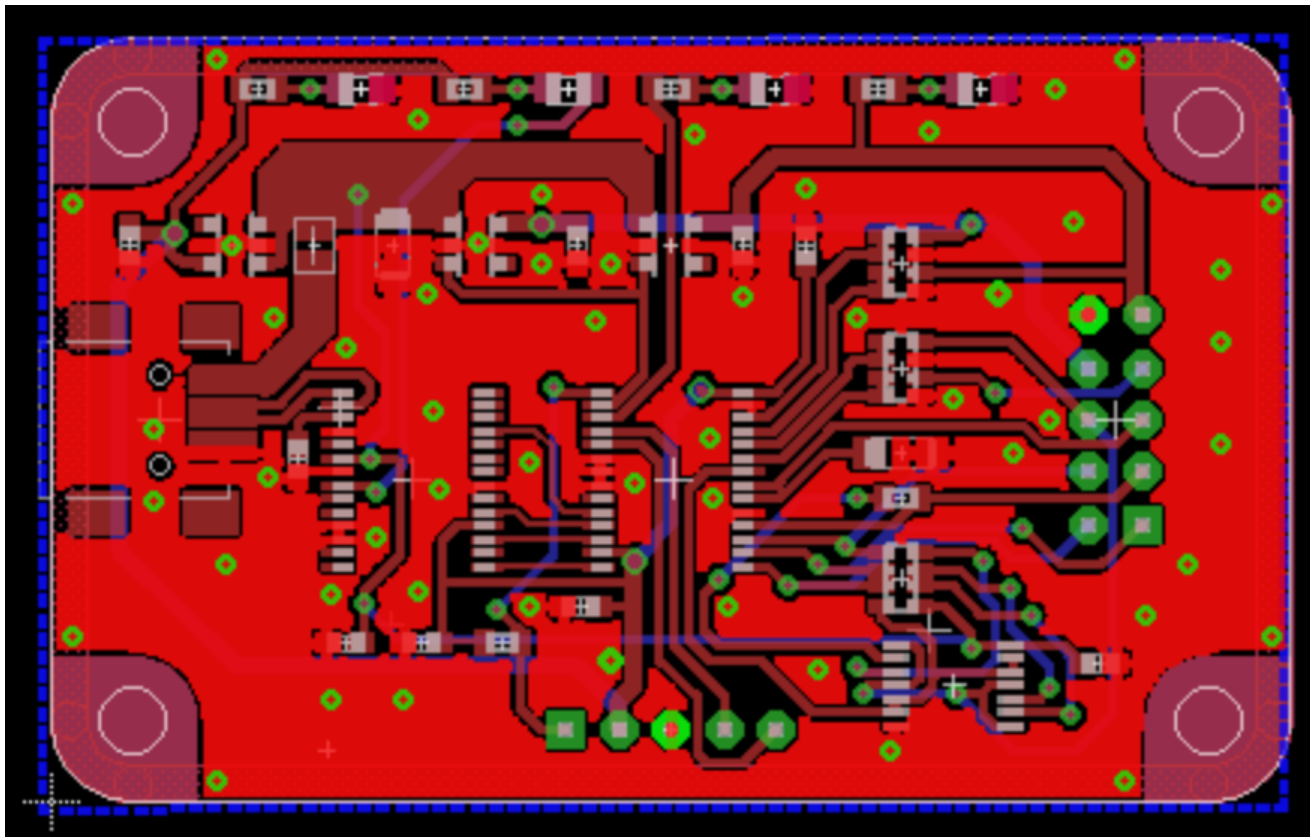
Width 3.5mil

Minimum trace width



Polygons

It is used (in general) when the designer wants to connect more than two objects on the same net (electrical reference).



Polygons

Polygons can be connected to component in several ways

Direct
Connection

Thermal Relief
Connection

Polygons

Polygons are always better for

- Power nets
- Ground nets
- High frequency nets
- Actually anything

Polygons

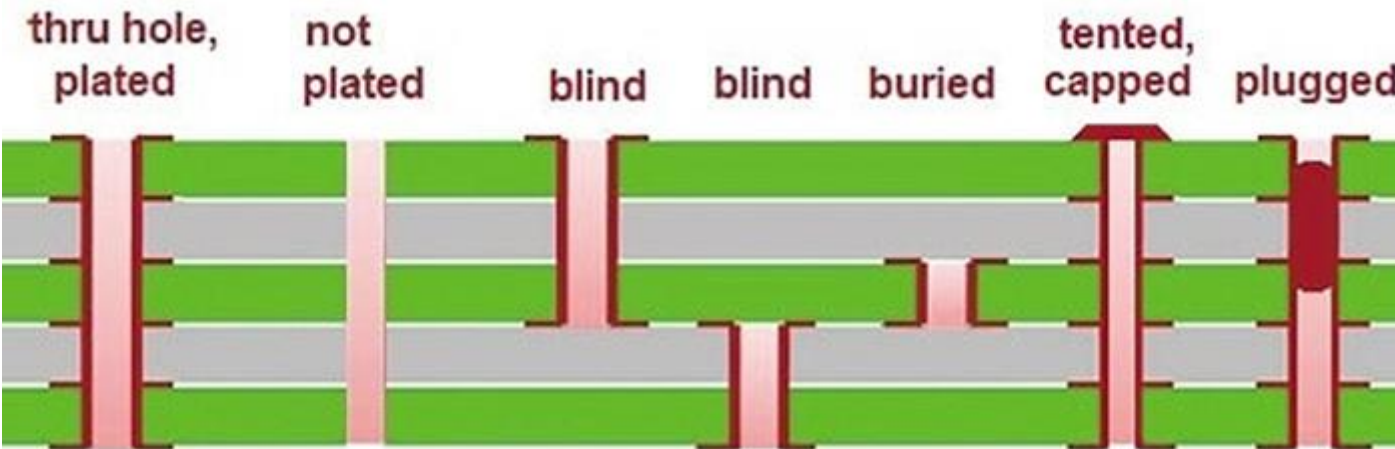
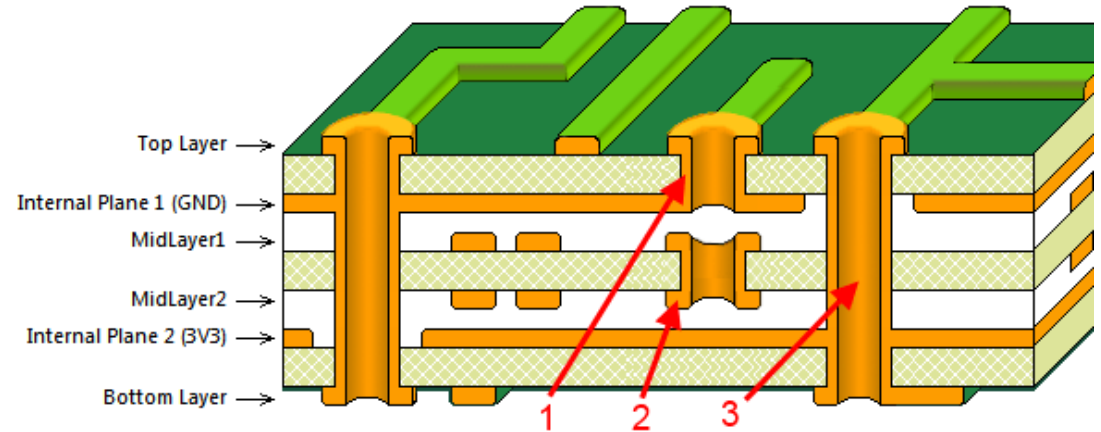
Polygons have priorities

Prior



Vias

Vias are used for connecting different conductive layers



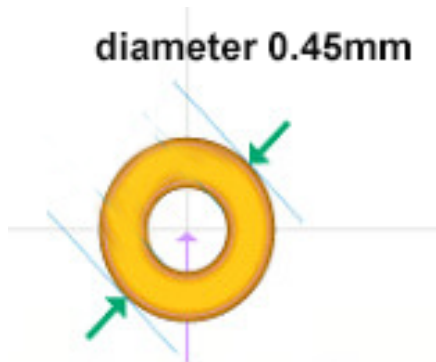
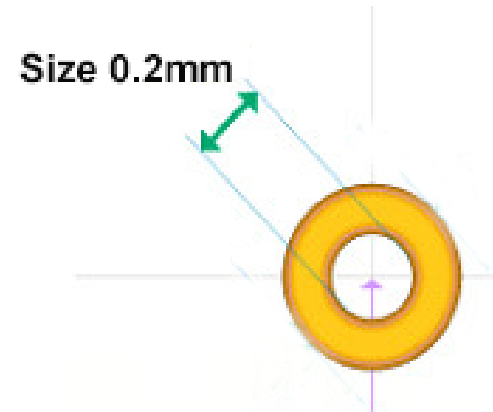
Vias

It might be required to use via array to have a low resistance current path or thermal path.

Vias

Check the capabilities of the manufacturer

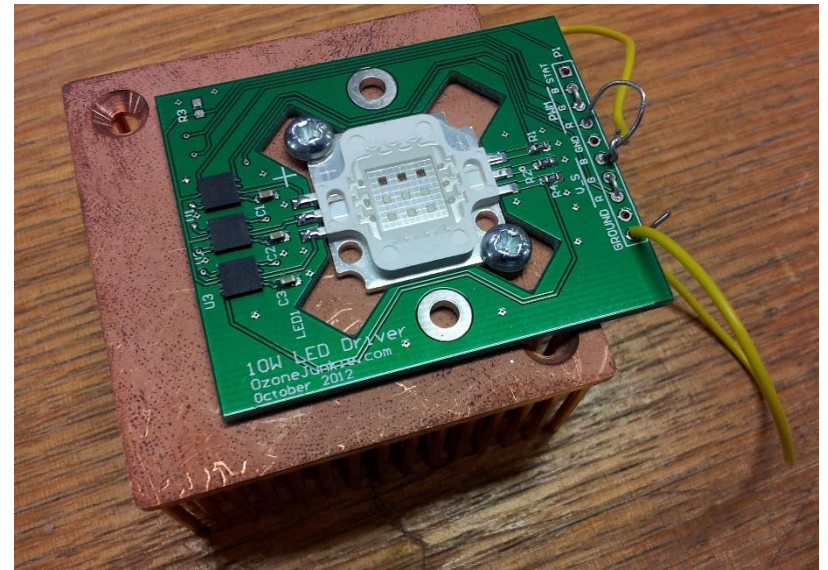
Minimum hole size



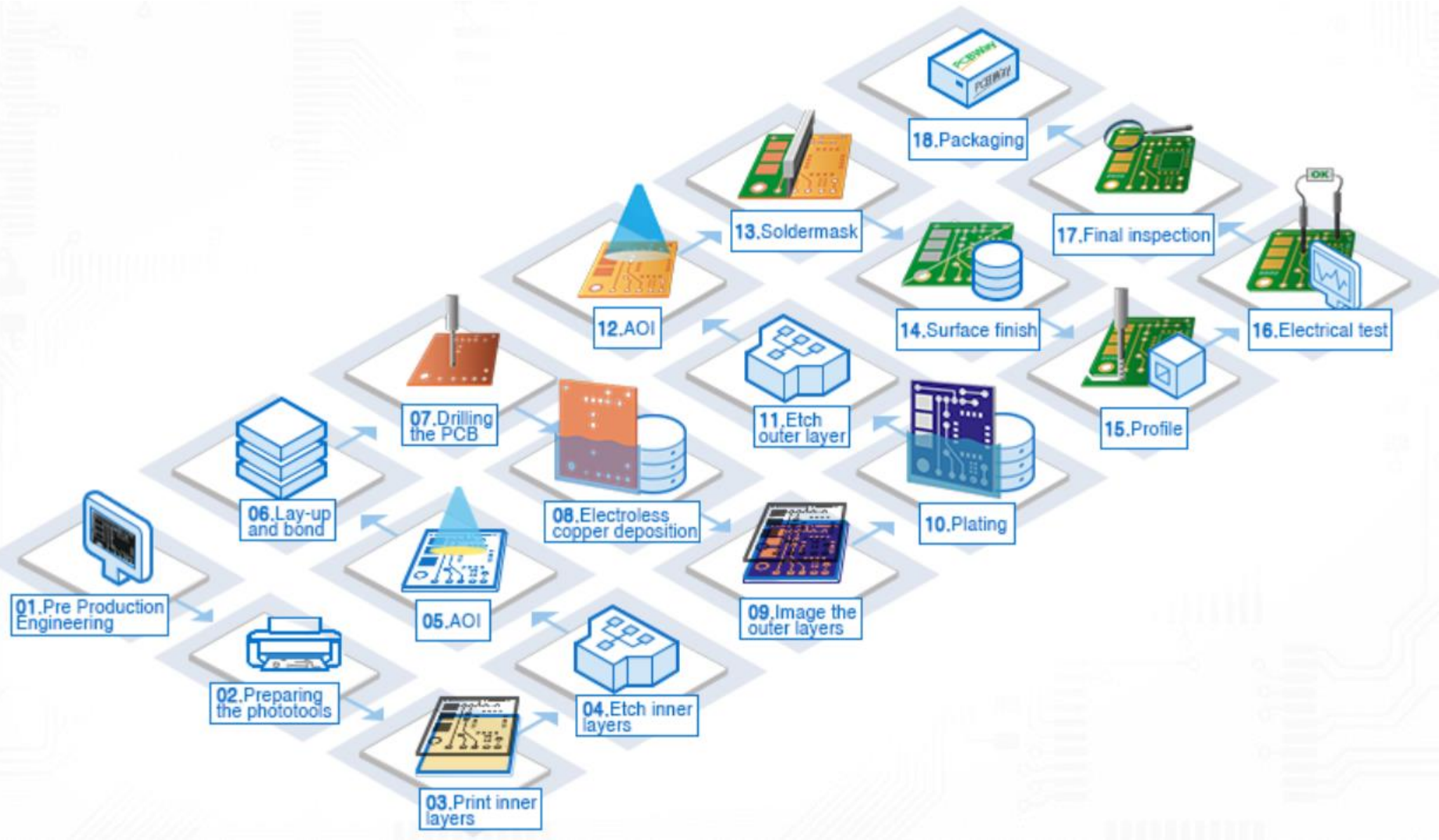
Minimum via diameter

Board Cutouts

Cutouts are used to have custom shapes

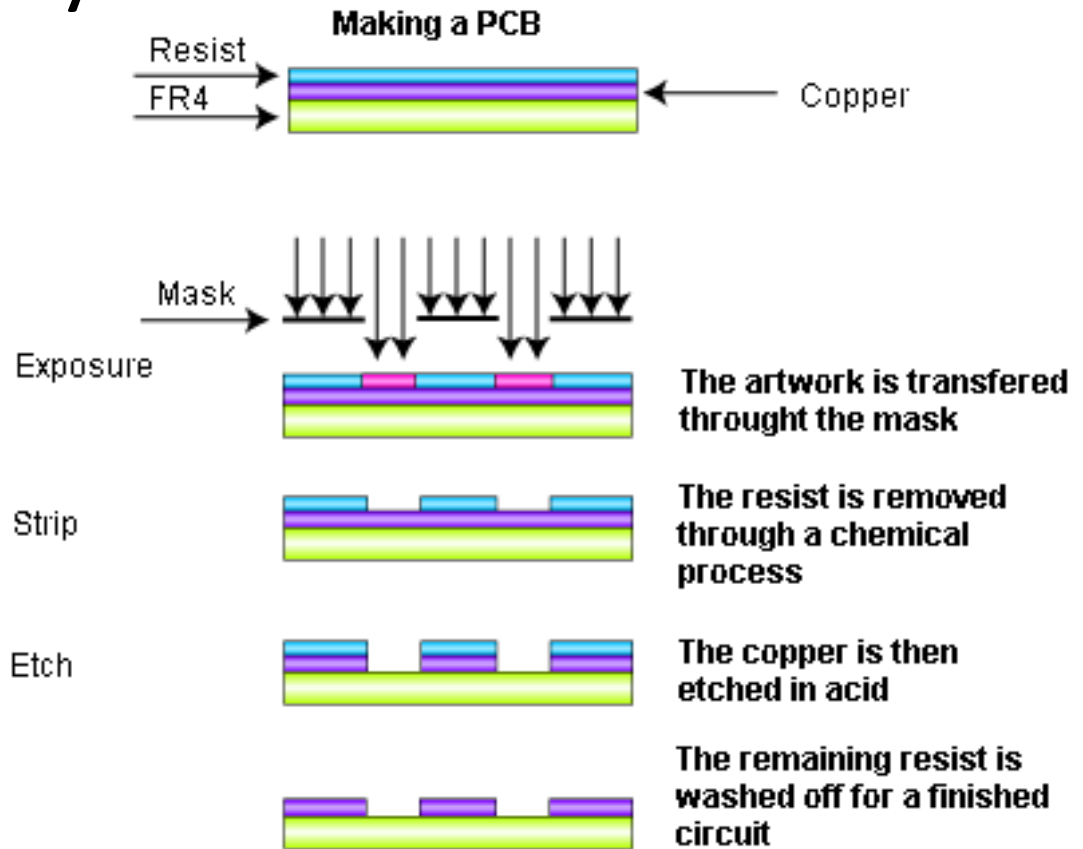


PCB Manufacturing Steps



PCB Manufacturing Steps

- Preparing the phototools
- Lytography
- Etching inner layers



PCB Manufacturing Steps

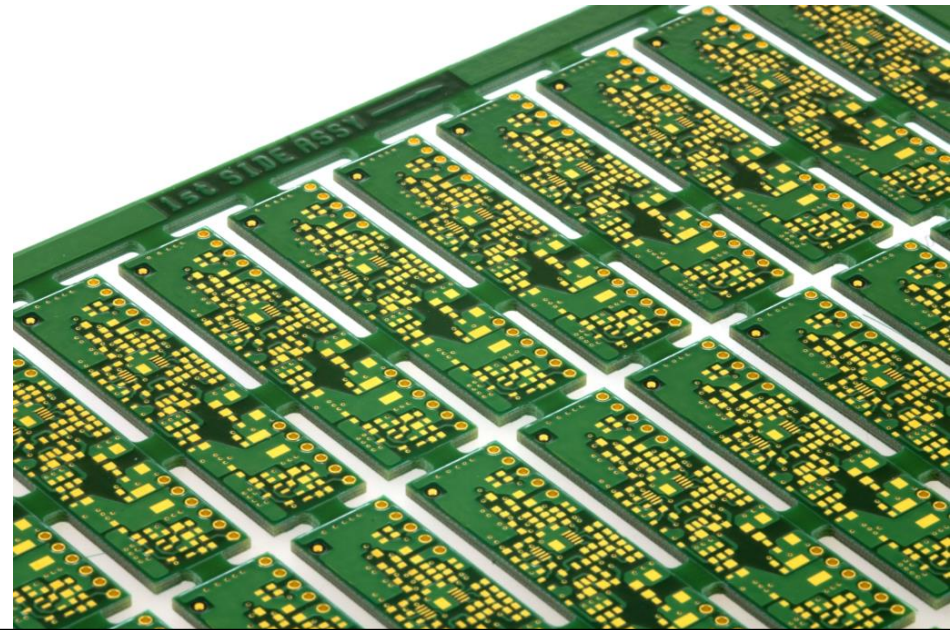
- Optical inspection
- Bonding inner layers
- Drilling inner layers



- Copper deposition inside holes for inner layers
- Plating copper for outer circuitry
- Etching outer layers
- Optical inspection

PCB Manufacturing Steps

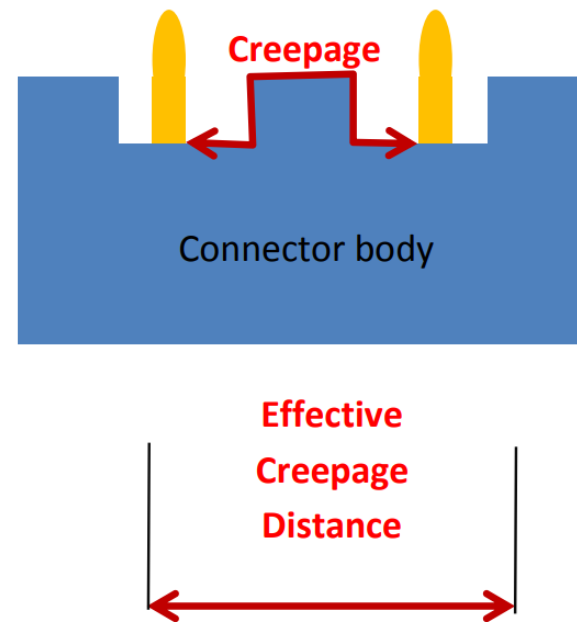
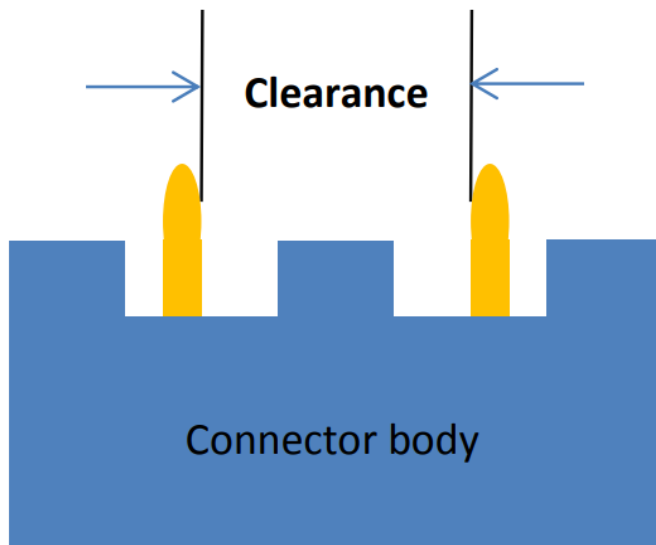
- Soldermask
- Surface finish: ENIG, HASL etc.
- Preparing the panel
- Electrical tests
- Final Inspection



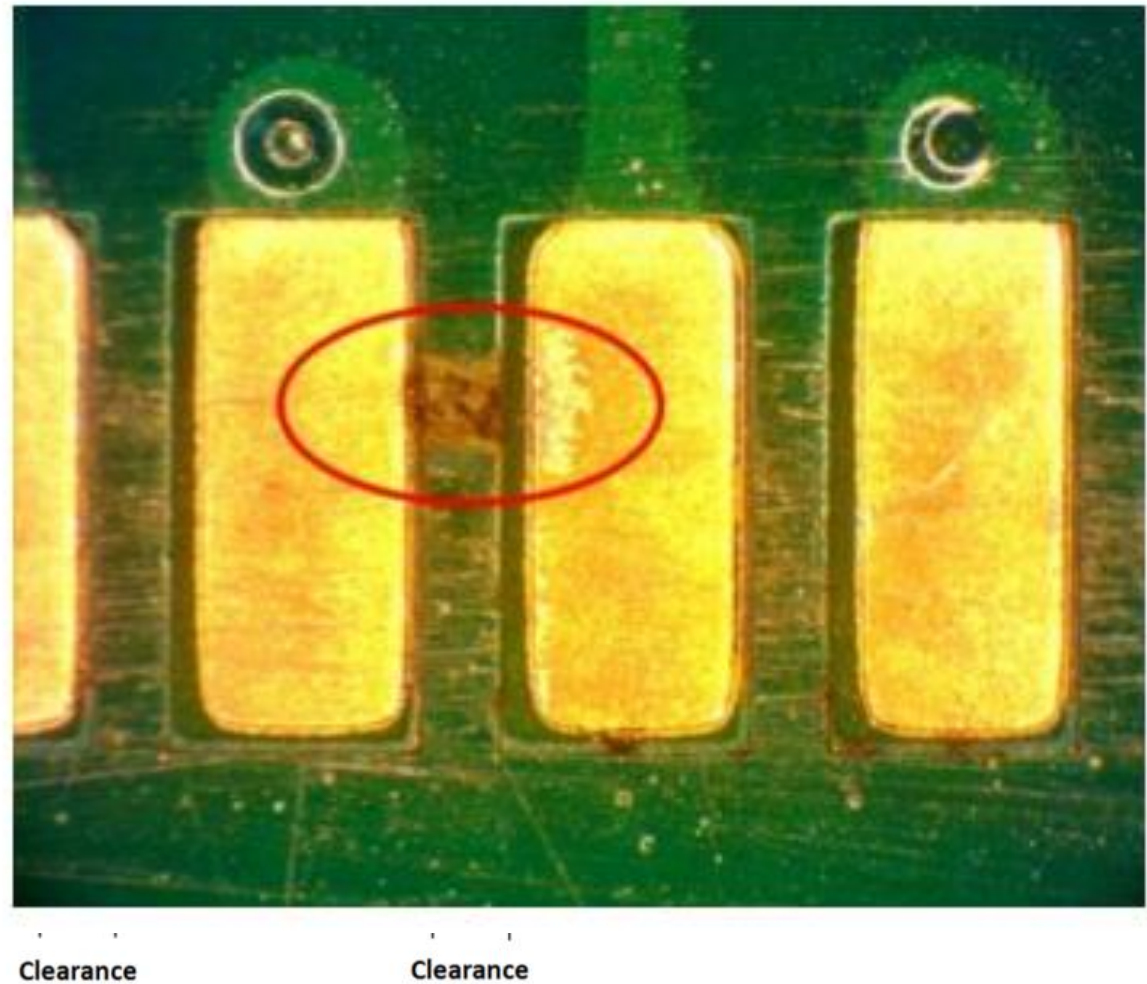
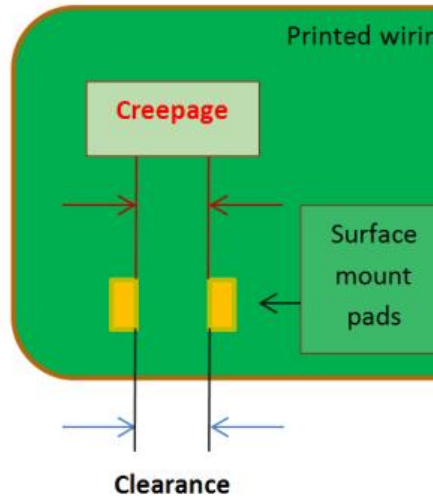
Concepts

Clearance

Clearance can be divided in two parts: electrical and mechanical.



Clearance



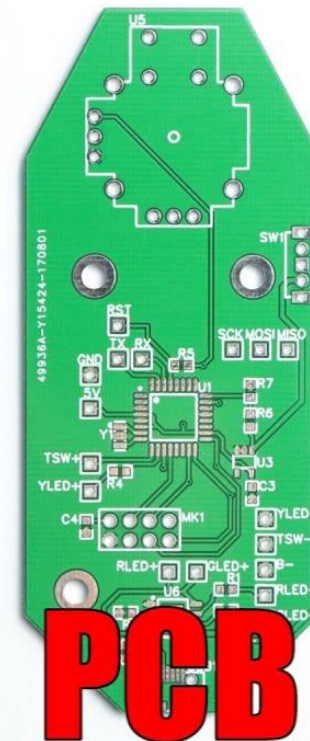
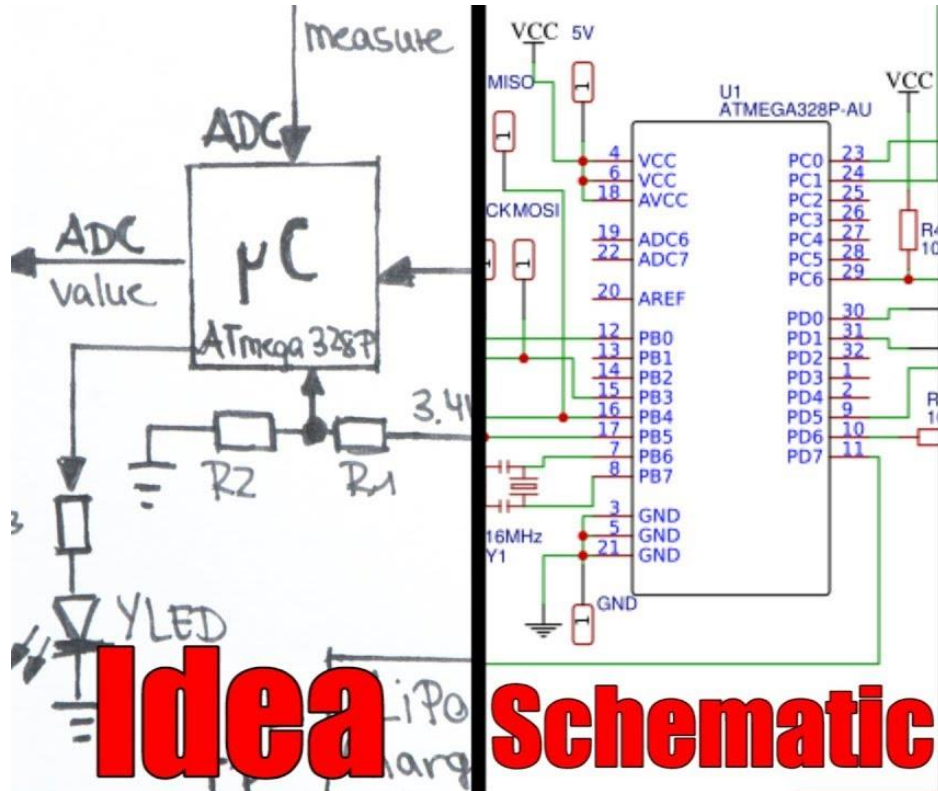
Clearance

Mechanical clearance is the distance between the bodies of two objects.



Schematic and Layout Design

All PCB design programs have two design concepts which are schematic design and layout design.

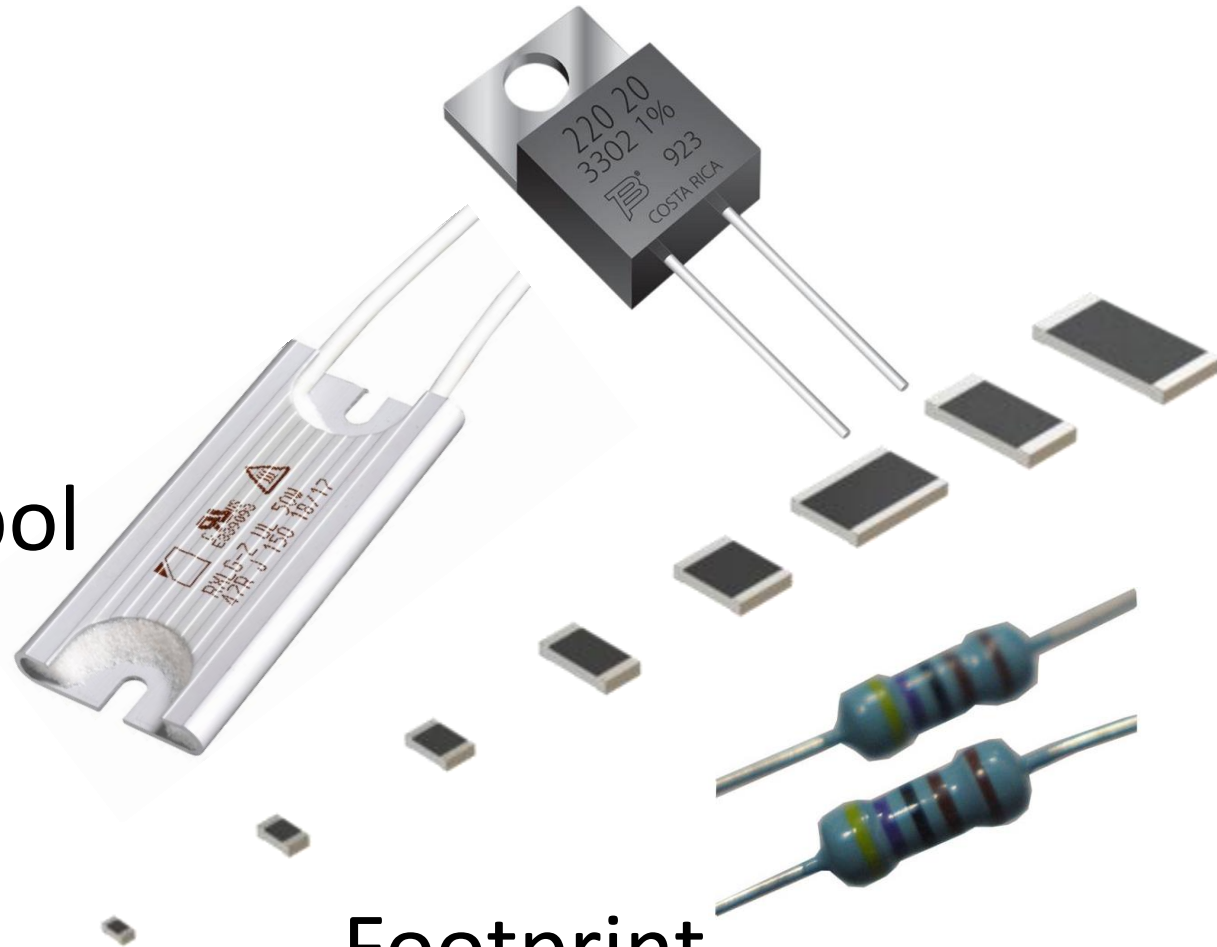


Schematic and Layout Design

All components have two part



Schematic Symbol



Footprint

Library

A library object
should be created
for a component
before placing it
on the board

Library

Embed more information as possible as
because you will forget them soon

- Designator
- Comment
- Description
- Link
- Cost



Library

You can either access the library document online or you can create it

- Check the manufacturer website
- Check open-source webpages
[Snapeda.com](https://www.snapeda.com) & [3dcontentcentral.com](https://www.3dcontentcentral.com)
- Create component using wizards
- Create component by hand

Rules

All generic constraints can be defined as a rule and the PCB design program automatically follows them.

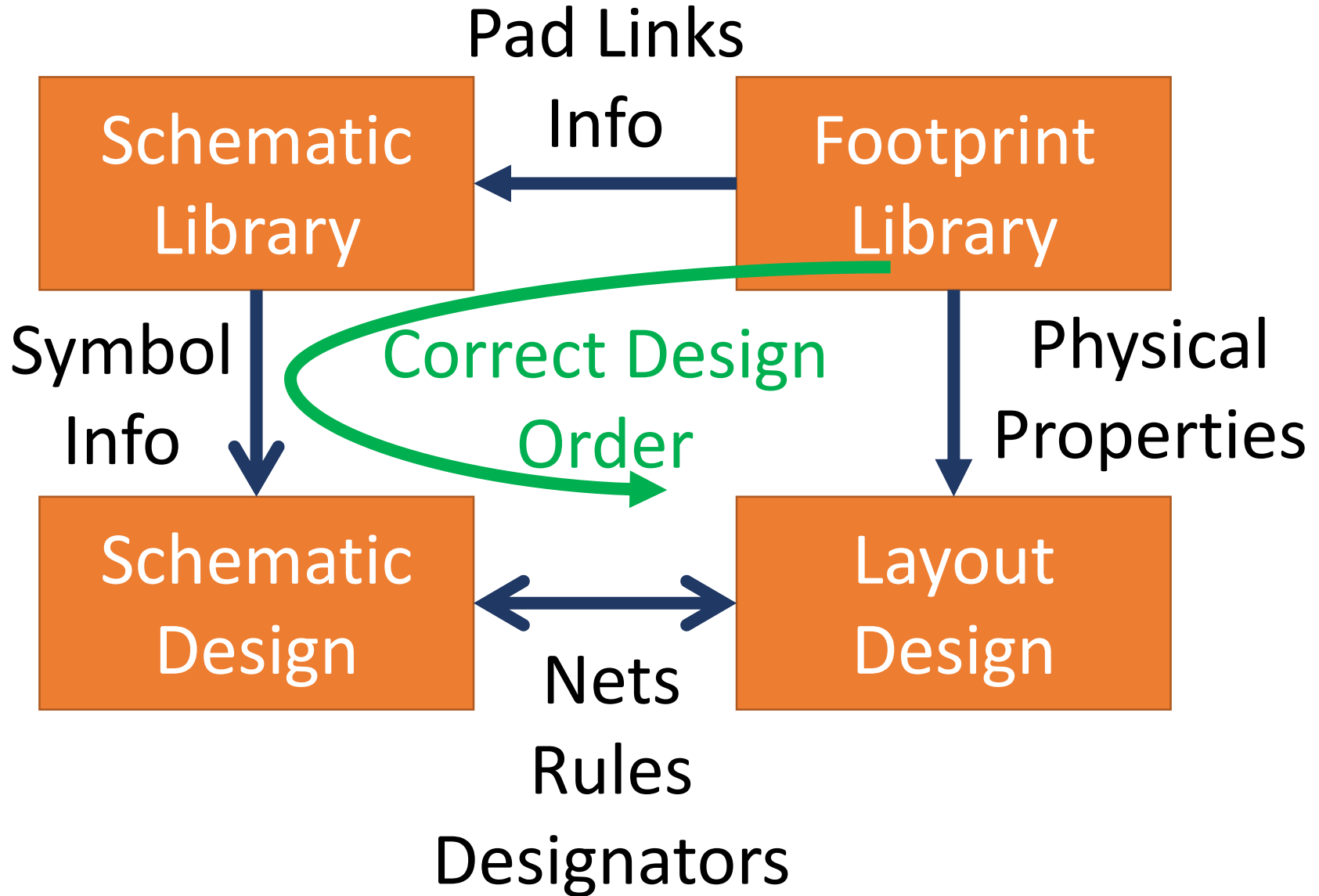
- Clearance
- Trace width
- Height
- Pad size
- Via size
- Much more thing

Classes

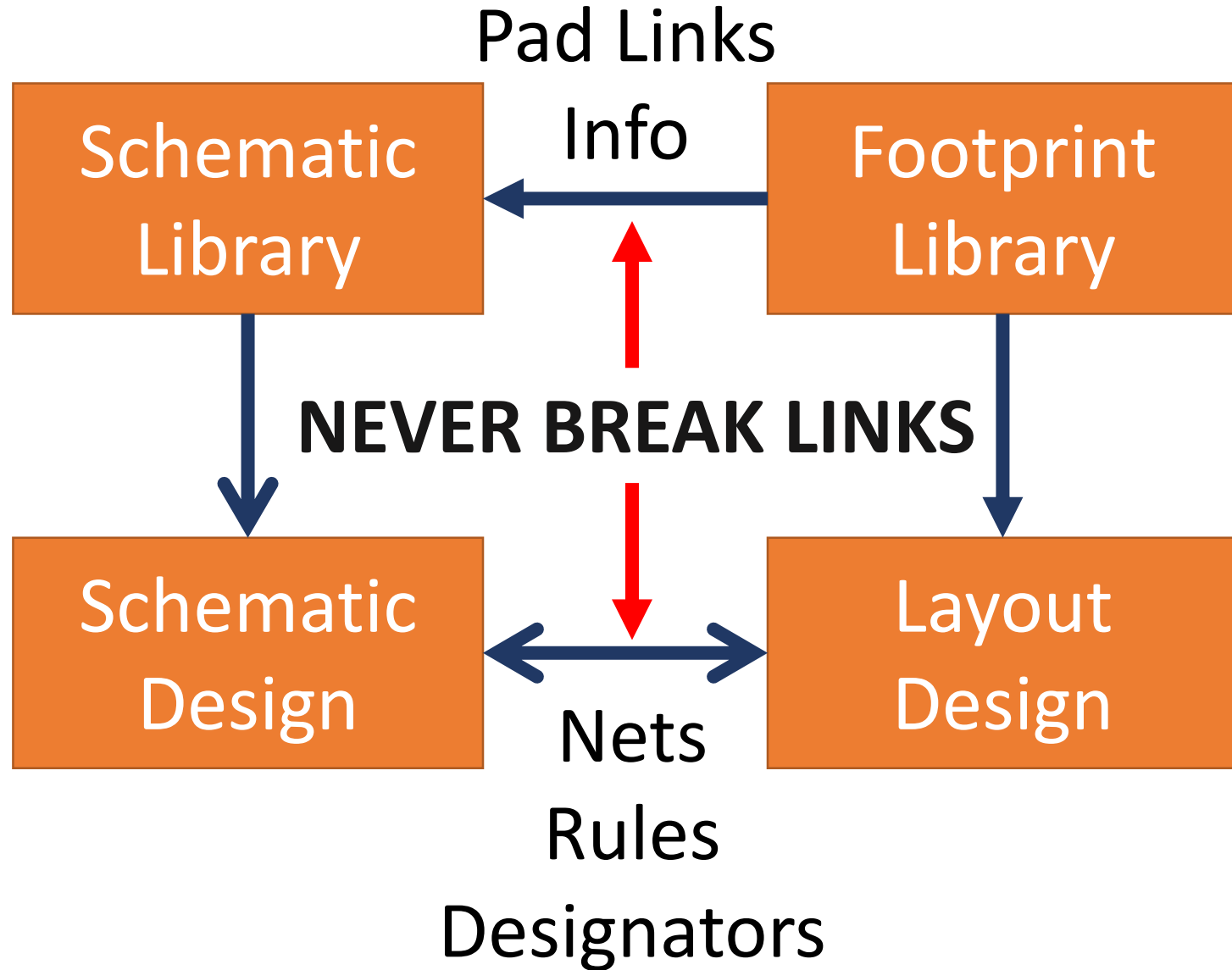
You can group objects and define rules between them.

- The minimum distance between copper and screw holes is 60 mils
- The nets of two isolated sides should be away from each other as minimum as 100 mils

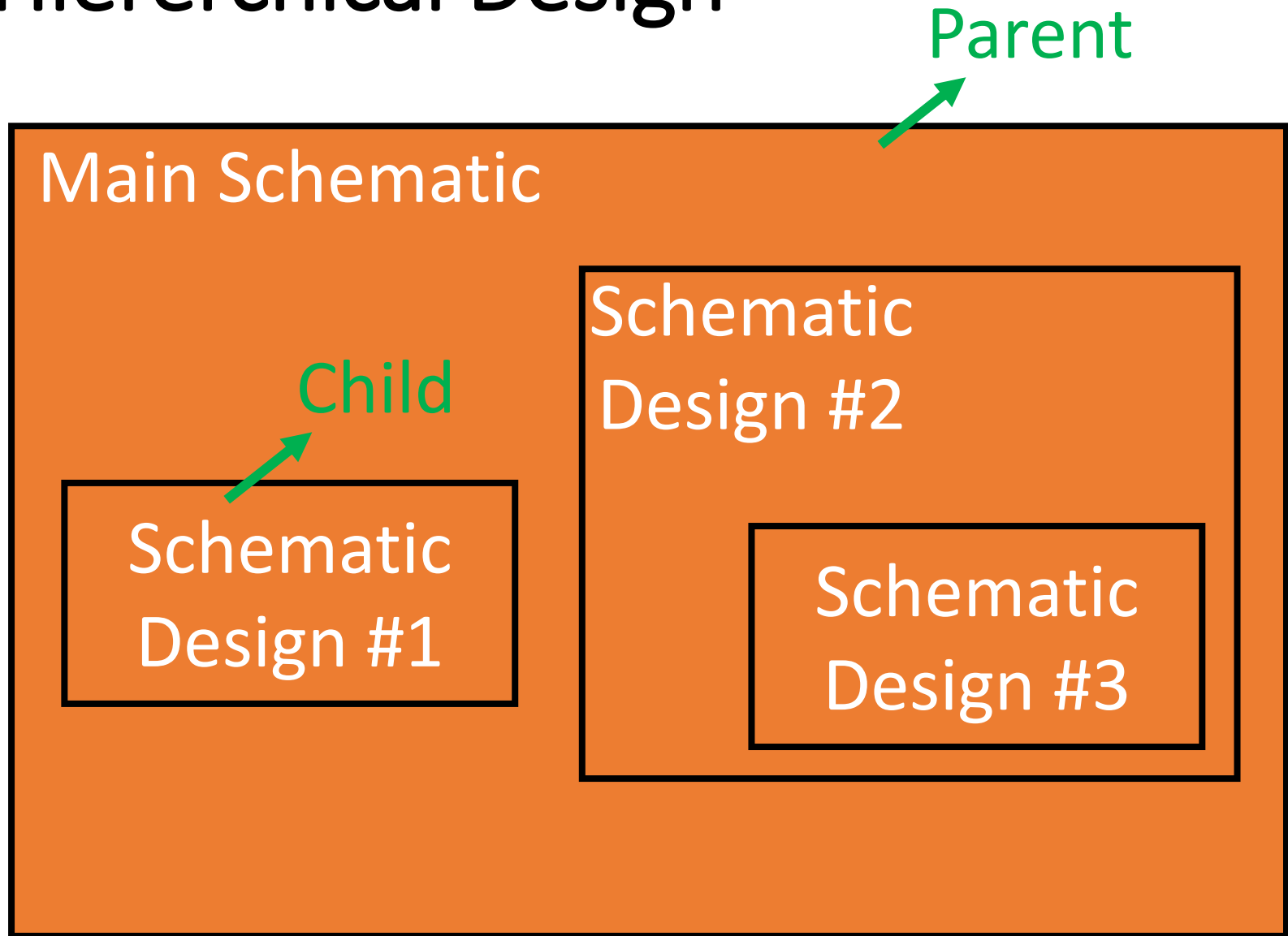
General Principle



General Principle



Hierarchical Design



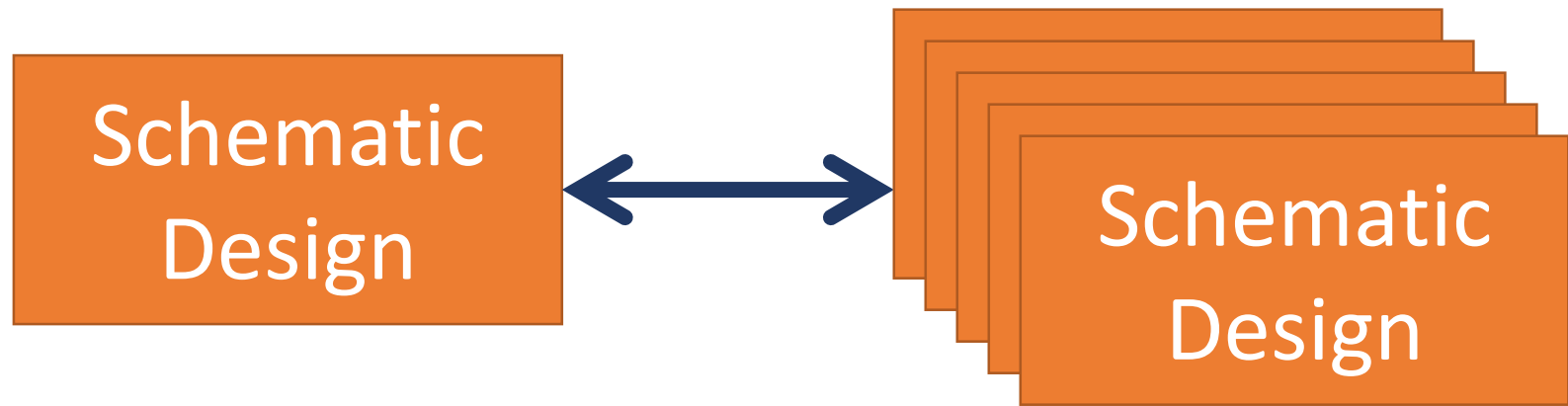
Hierarchical Design

Hierarchical design makes your project

- More understandable
- More organized
- Reusable
- Printable

It is strongly advised to utilize hierarchical design.

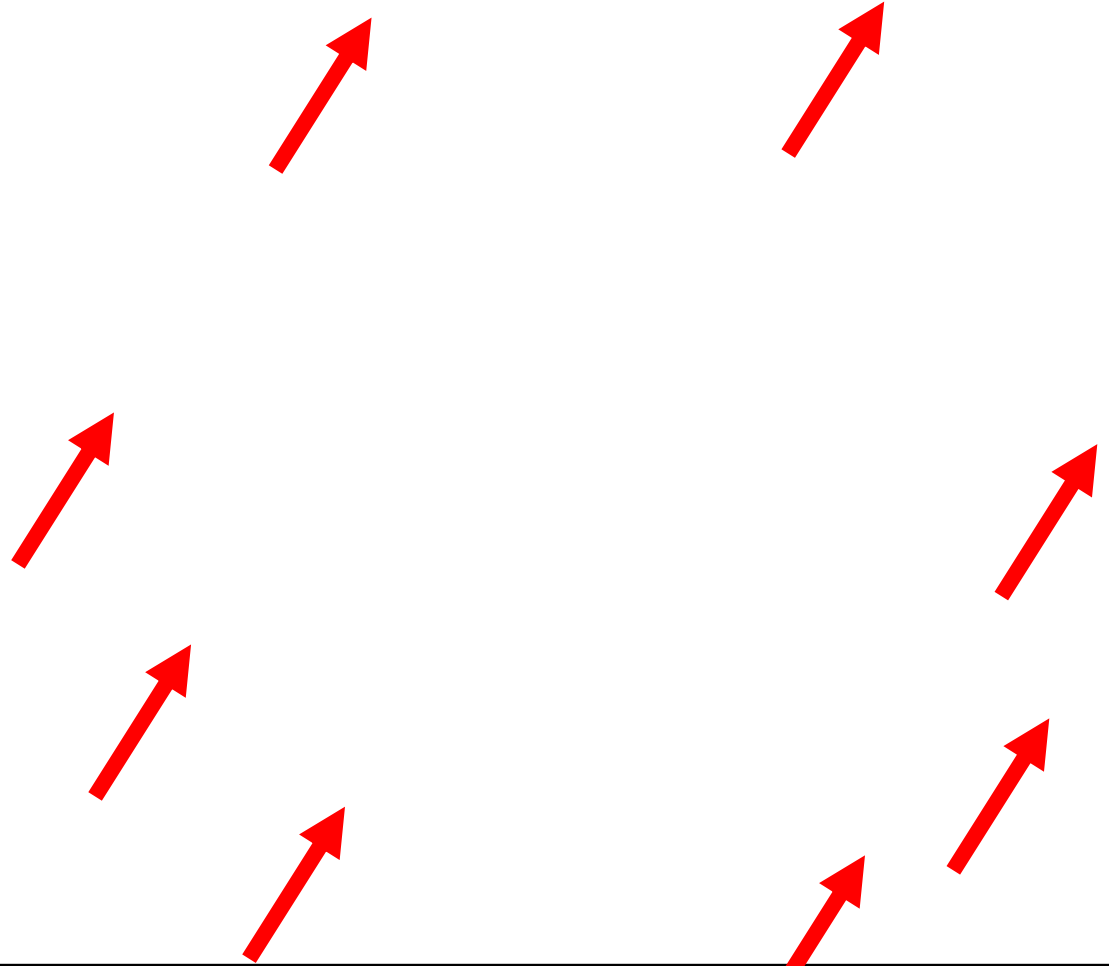
Multichannel Design



If it repeats itself, you don't have to design each separately.

Moreover, if you want to change something later, you don't have to apply it one by one.

Multichannel Design



Multiboard Design



If you need to verify your connectivity and mechanical integration, use multiboard design.

Readable PCB

In order to have a readable PCB, you should

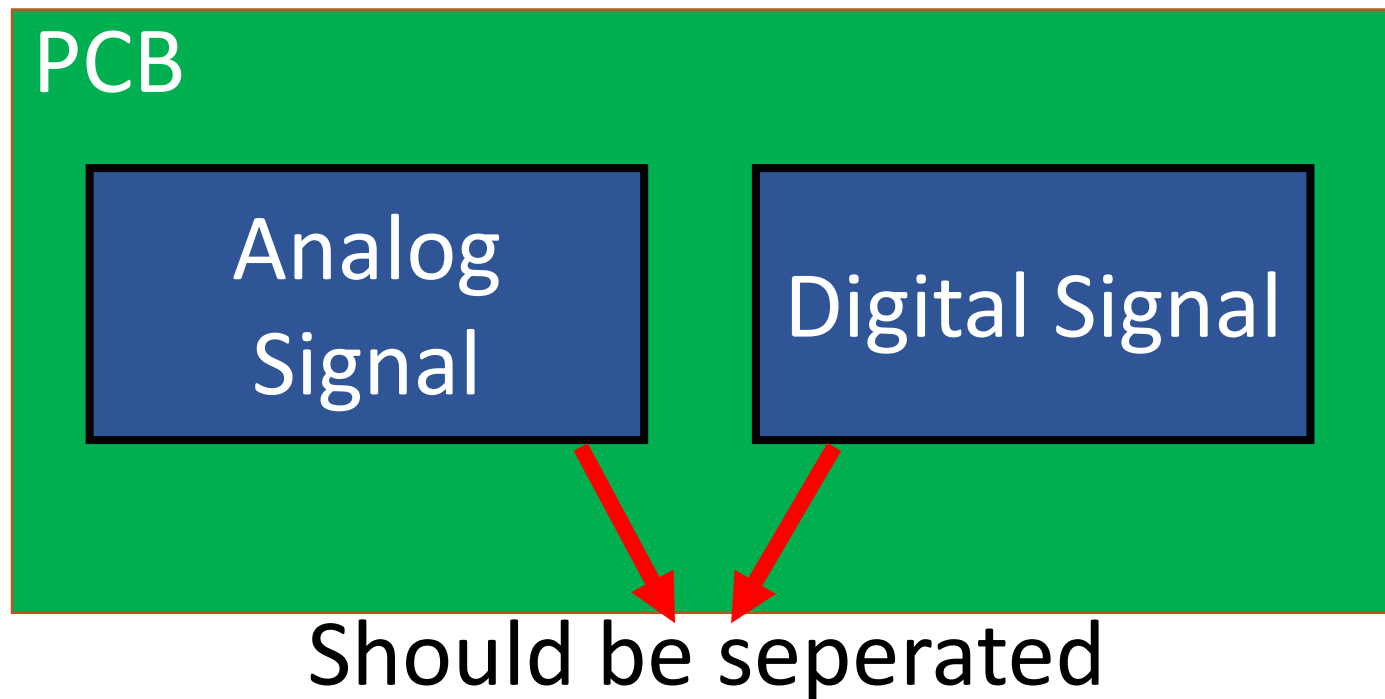
- Apply positional annotation
- Write designators
- Place instructions

Silk screen exists for this purpose.

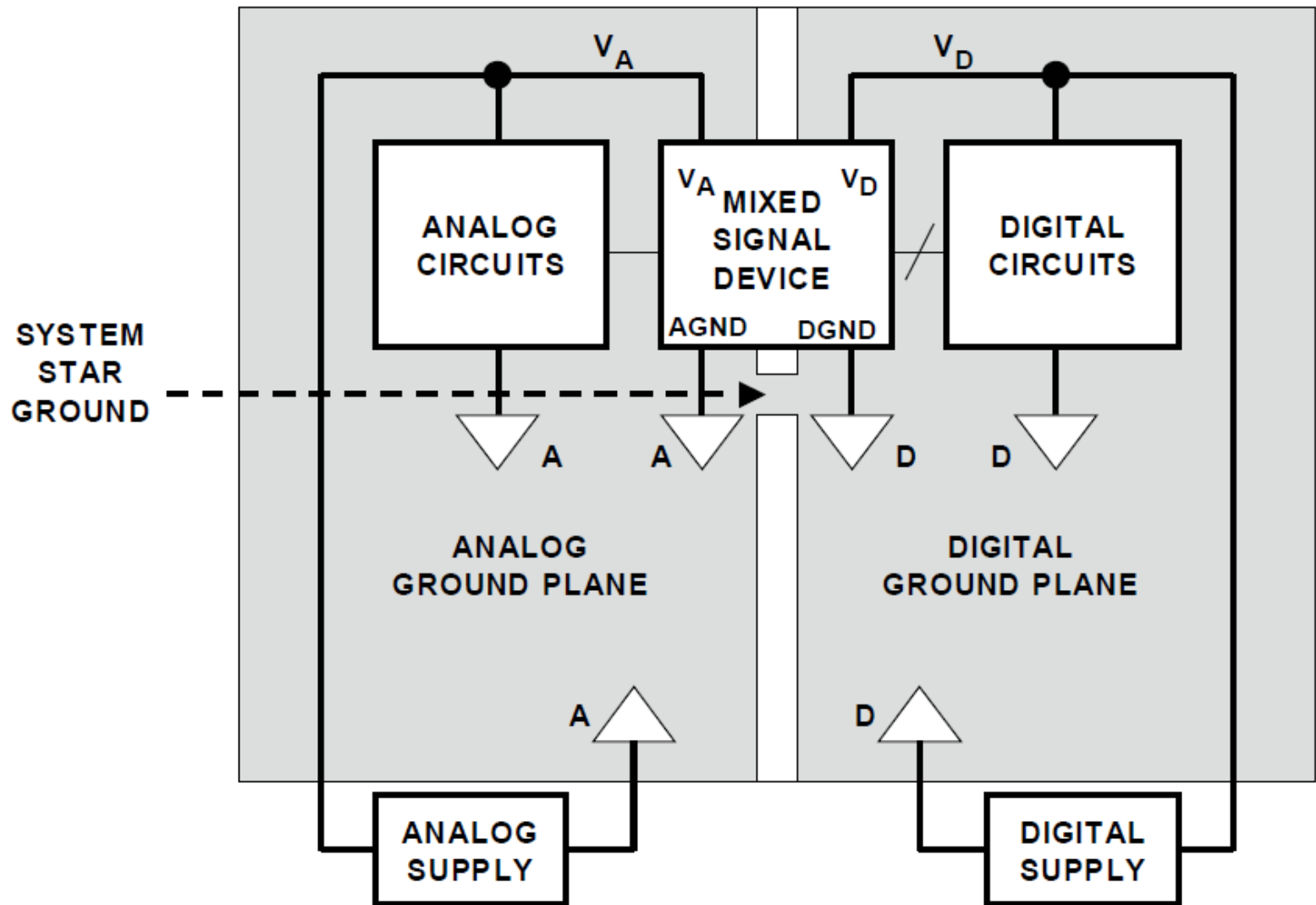
Specific Design Techniques

Mixed Signal Board

This is a board which includes digital signals, low level analog signals and/or high level analog signals, i.e. power nets.

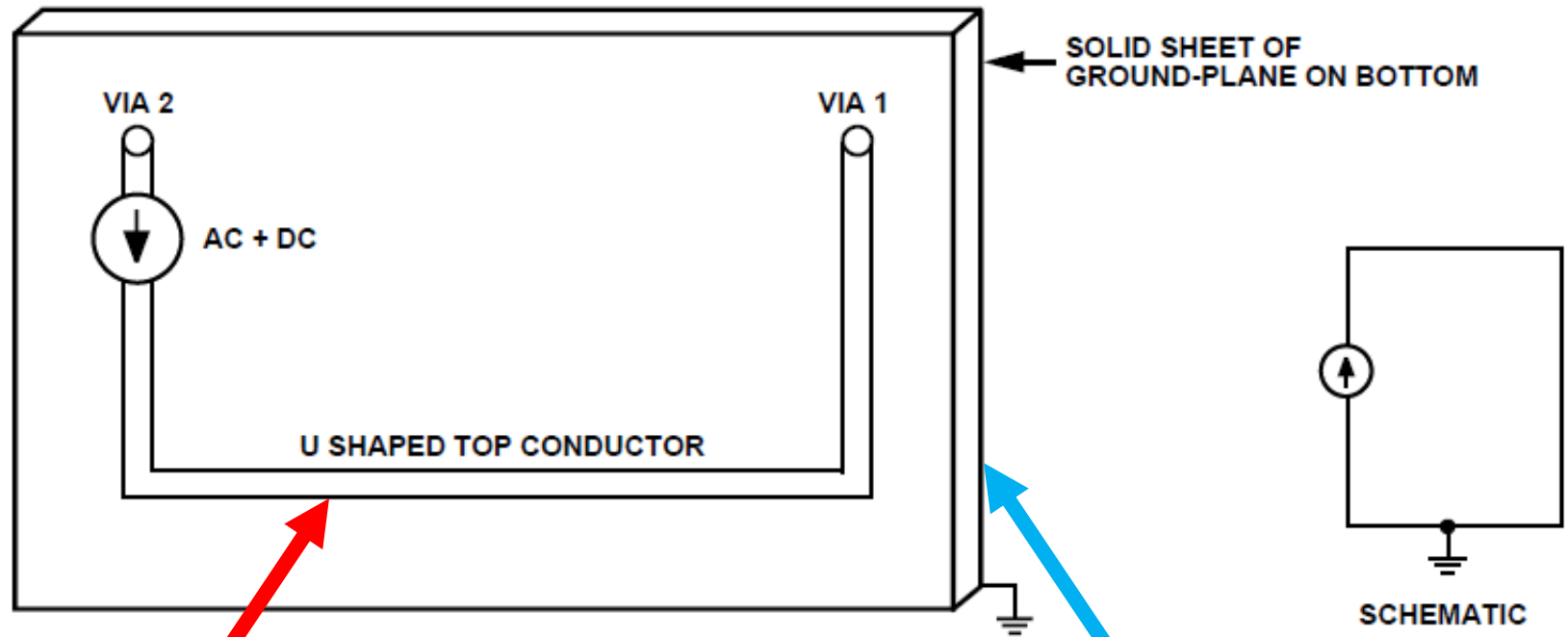


Mixed Signal Board



Mixed Signal Board

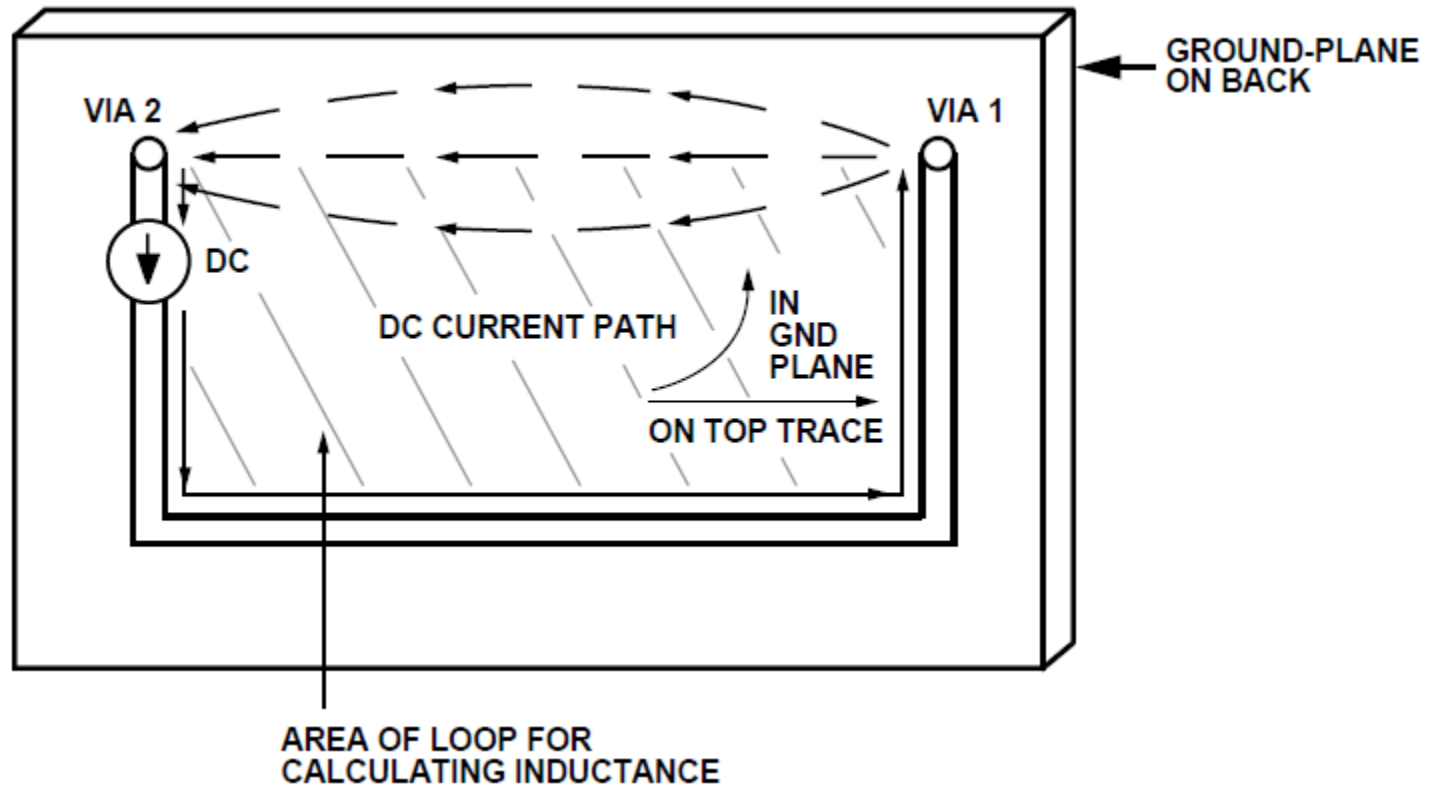
Grounding High Frequency Signal



Conductive path on
top layer

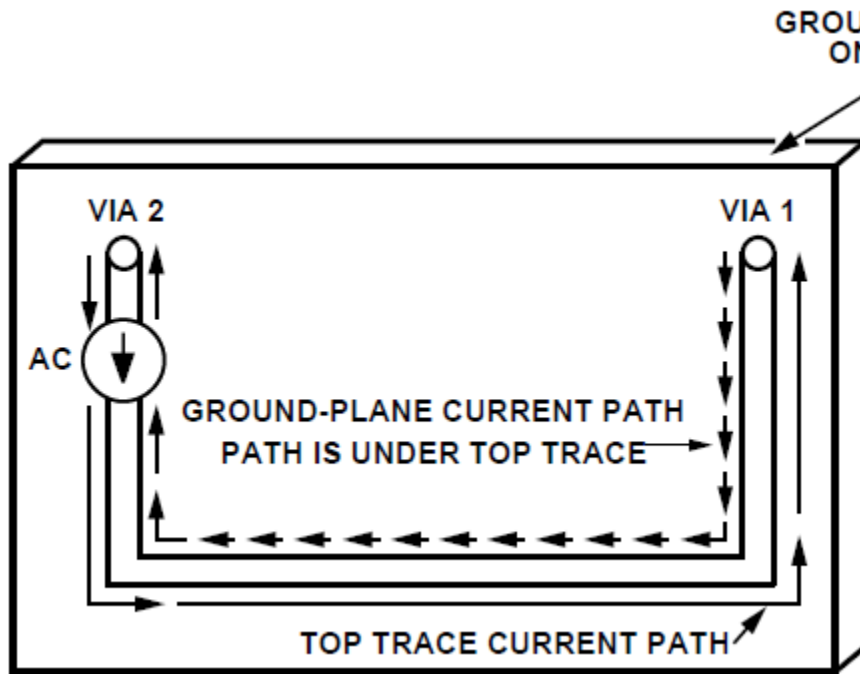
Solid polygon on
bottom layer

Grounding High Frequency Signal

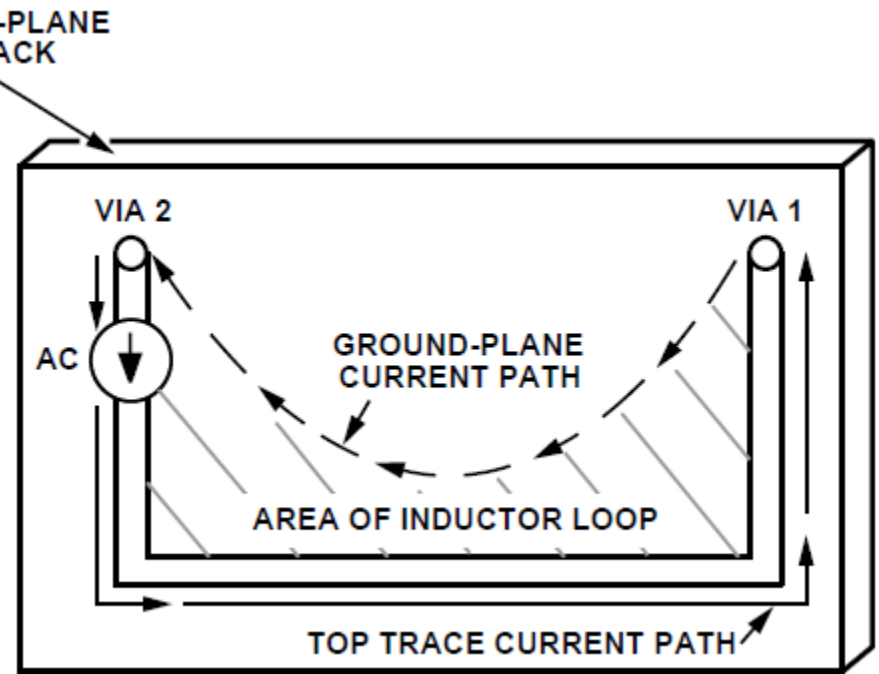


Main DC current flows through the minimum resistance path

Grounding High Frequency Signal



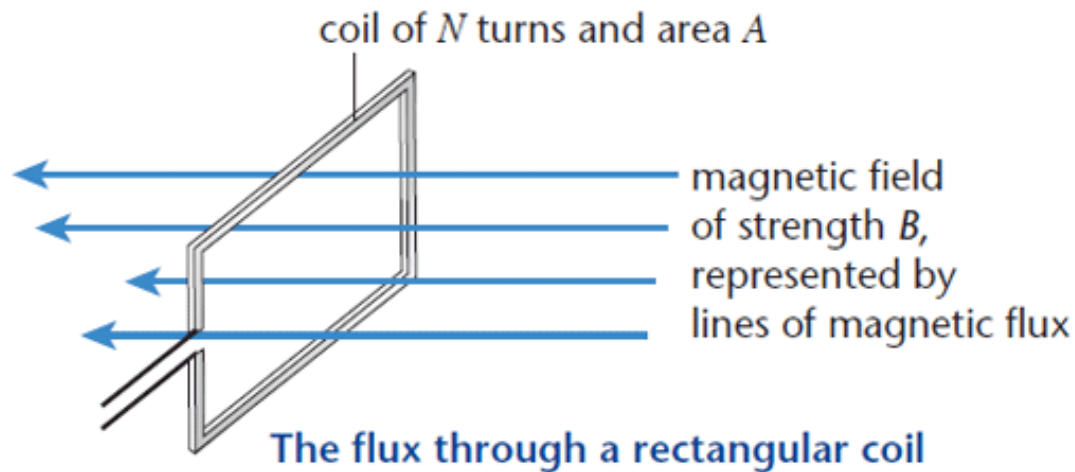
Ground plane
resistance is **ignored**



Ground plane resistance
is **not ignored**

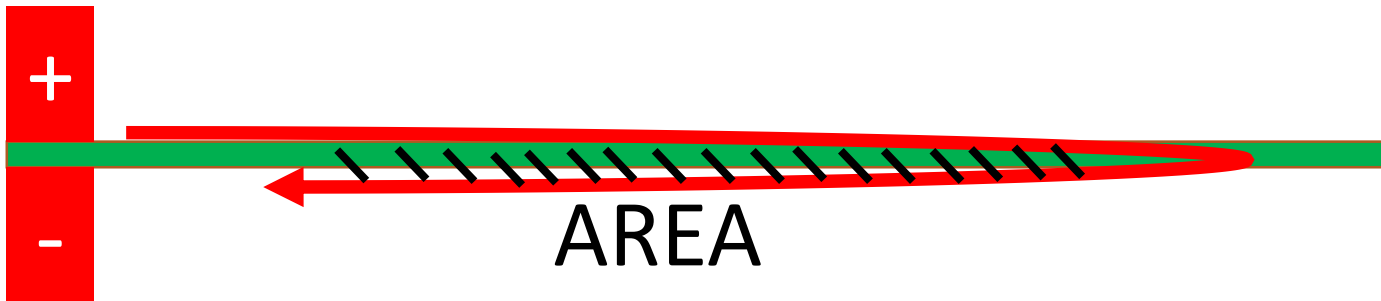
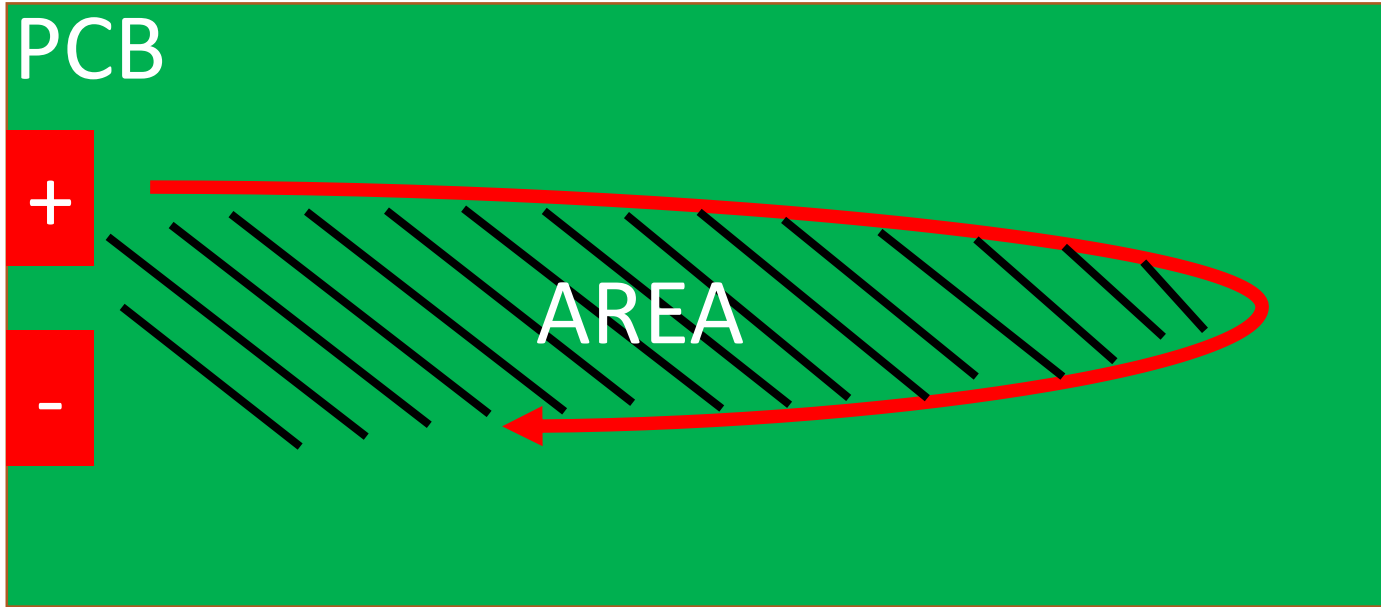
Grounding High Frequency Signal

Why is the minimum inductance is obtained for the paths on top of each other?



An AC current creates a magnetic field in a loop

Grounding High Frequency Signal



Grounding High Frequency Signal

A designer should be able to sense where the current flows and the designer should provide

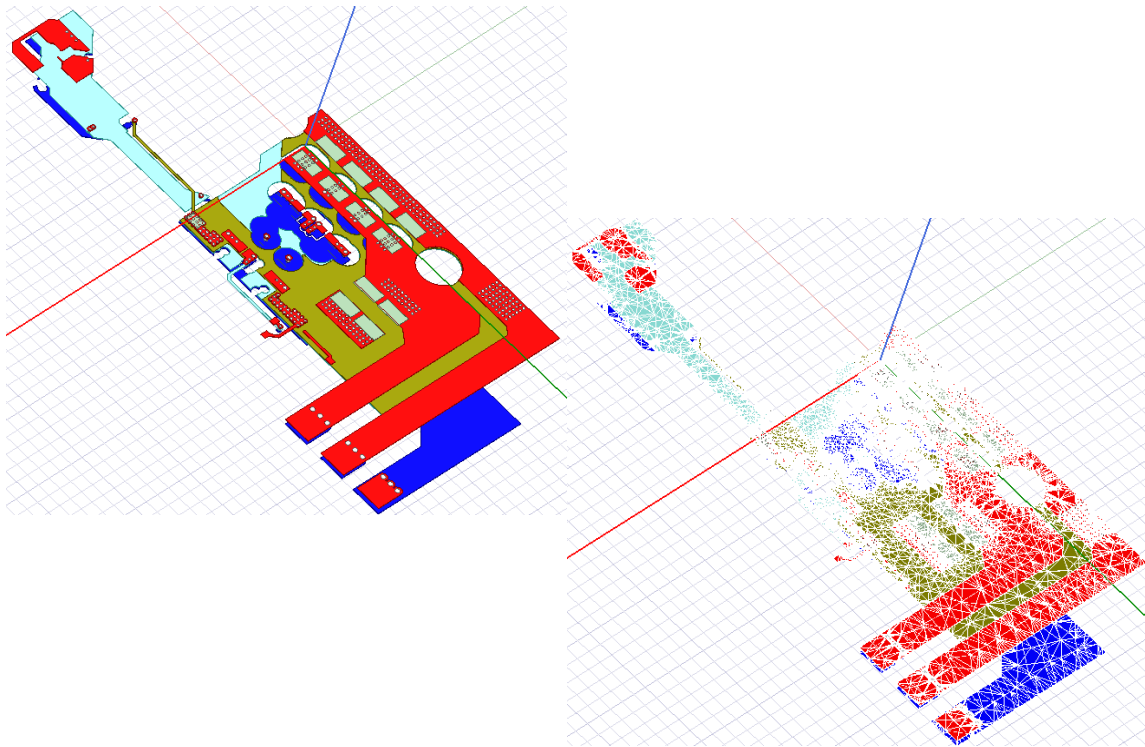
- Minimum resistance path for DC current
- Minimum inductance path for AC current



A switching transient current of a half-bridge is counted as a high frequency signal.

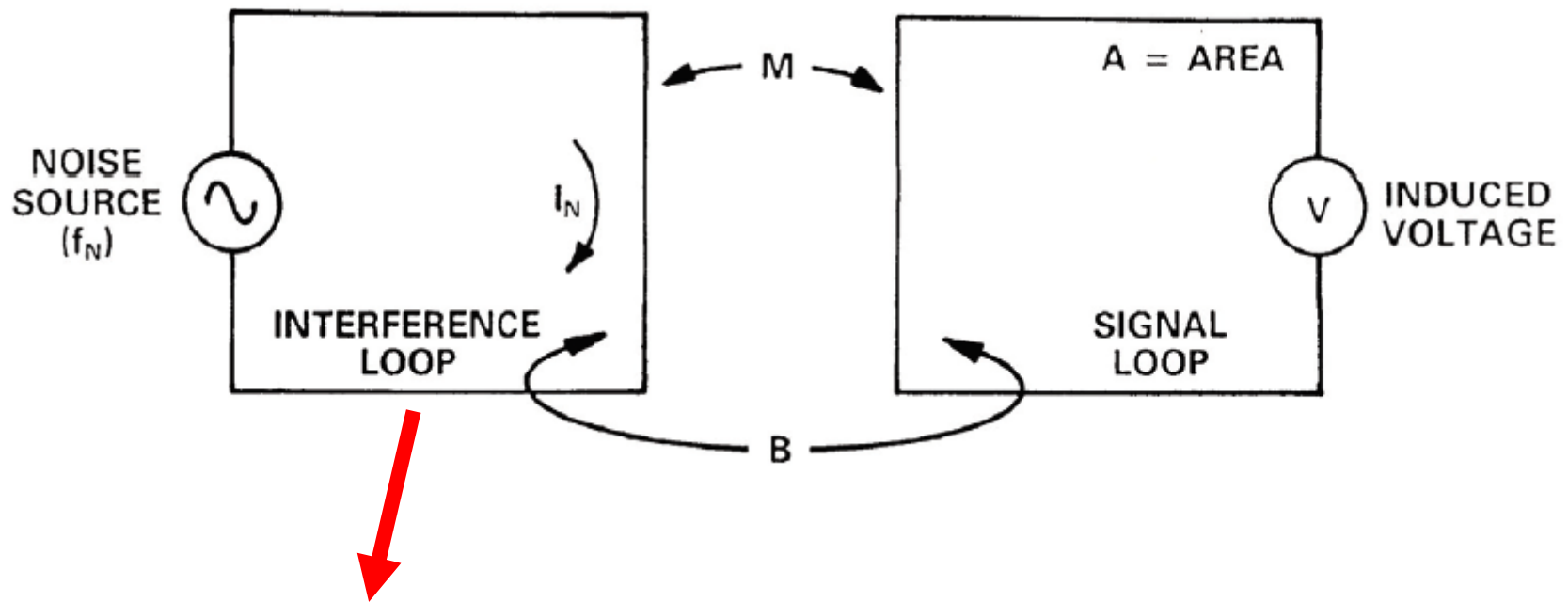
Grounding High Frequency Signal

Sometimes, it is not possible to easily estimate the correct path or you might need to find exact inductance/resistance.



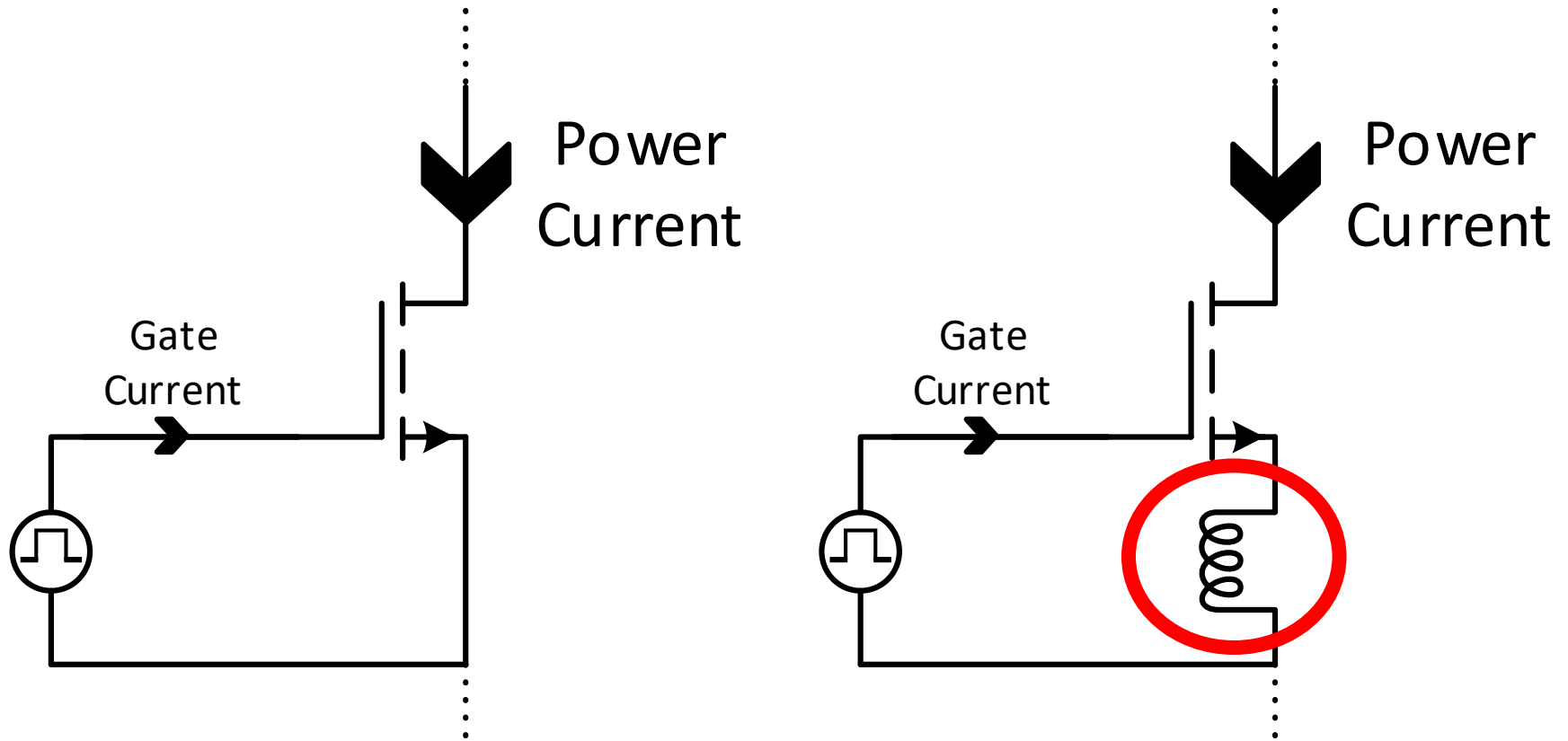
Use a FEM
analysis program
such as
Q3D Extractor or
COMSOL

Mutual Effects



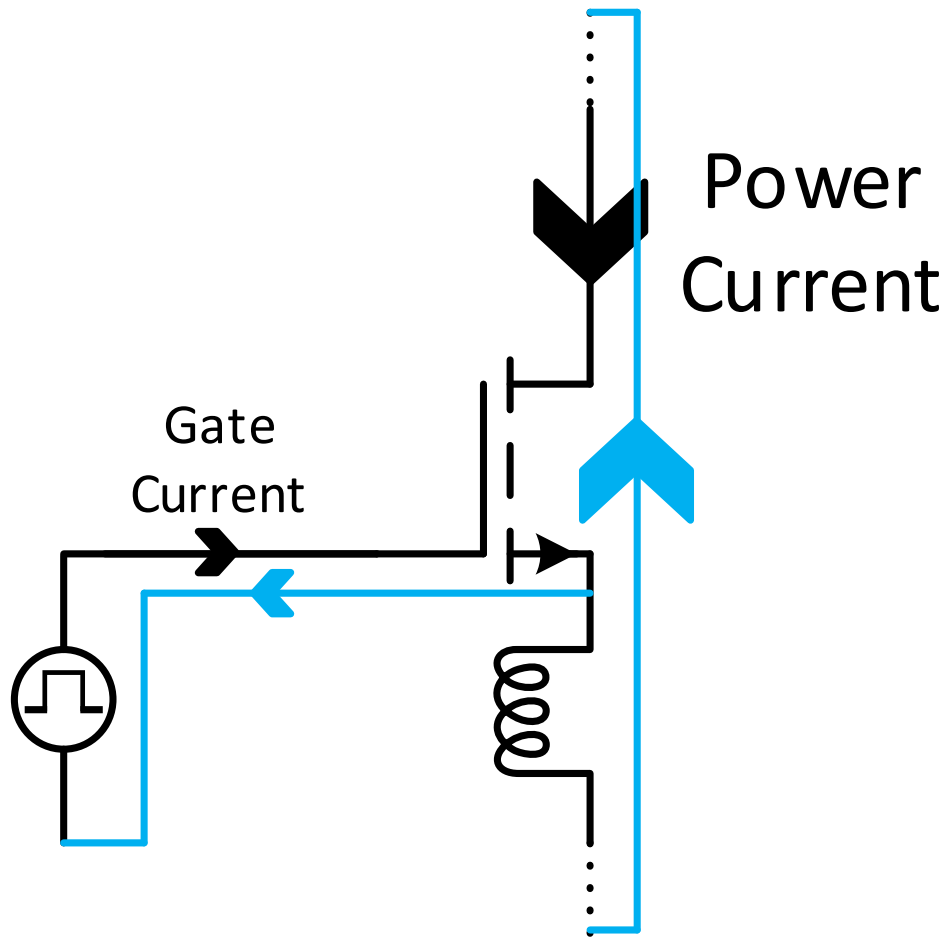
If this noise gets stronger, it will disturb all analog signals, digital signals and other power signals as well.

Gate Loop and Power Loop



What is the source for this inductance?

Gate Loop and Power Loop

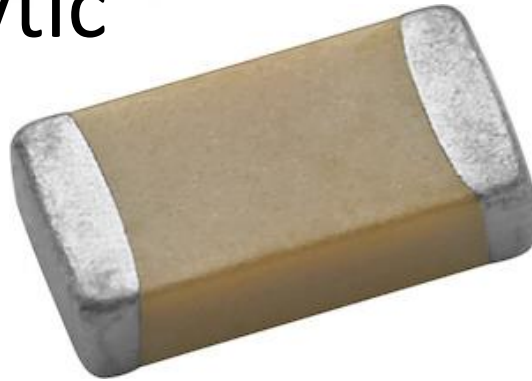


- Minimize all loops as possible as
- Seperate the loops from each other

Using Capacitors Effective



Electrolytic



Ceramic



Film

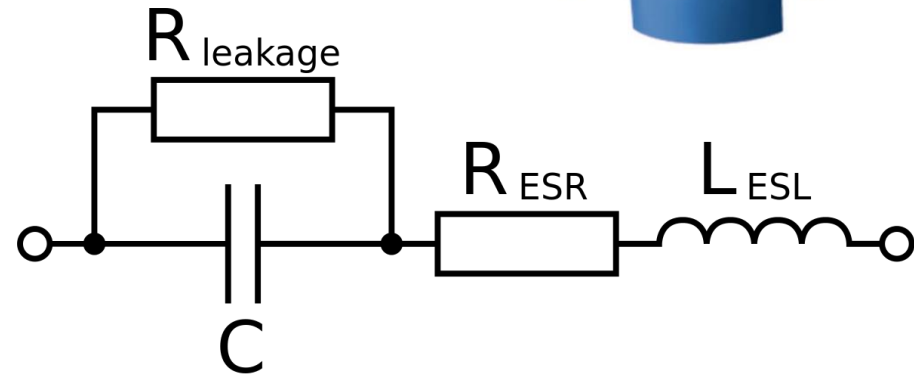
What can be the differences of these capacitors?

Using Capacitors Effective

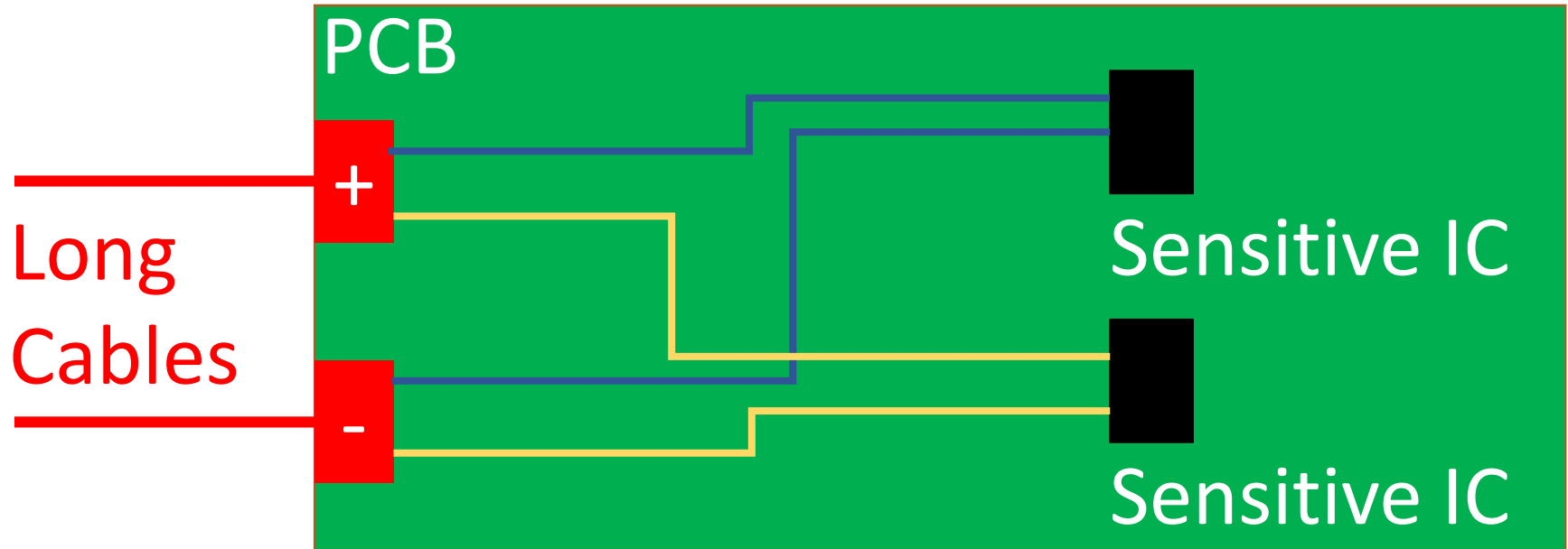
A capacitor is never just a capacitor



- Capacitance density
- **Bandwidth**
- Temperature sensitivity
- Current handling capability
- Post-failure situation
- Lifetime

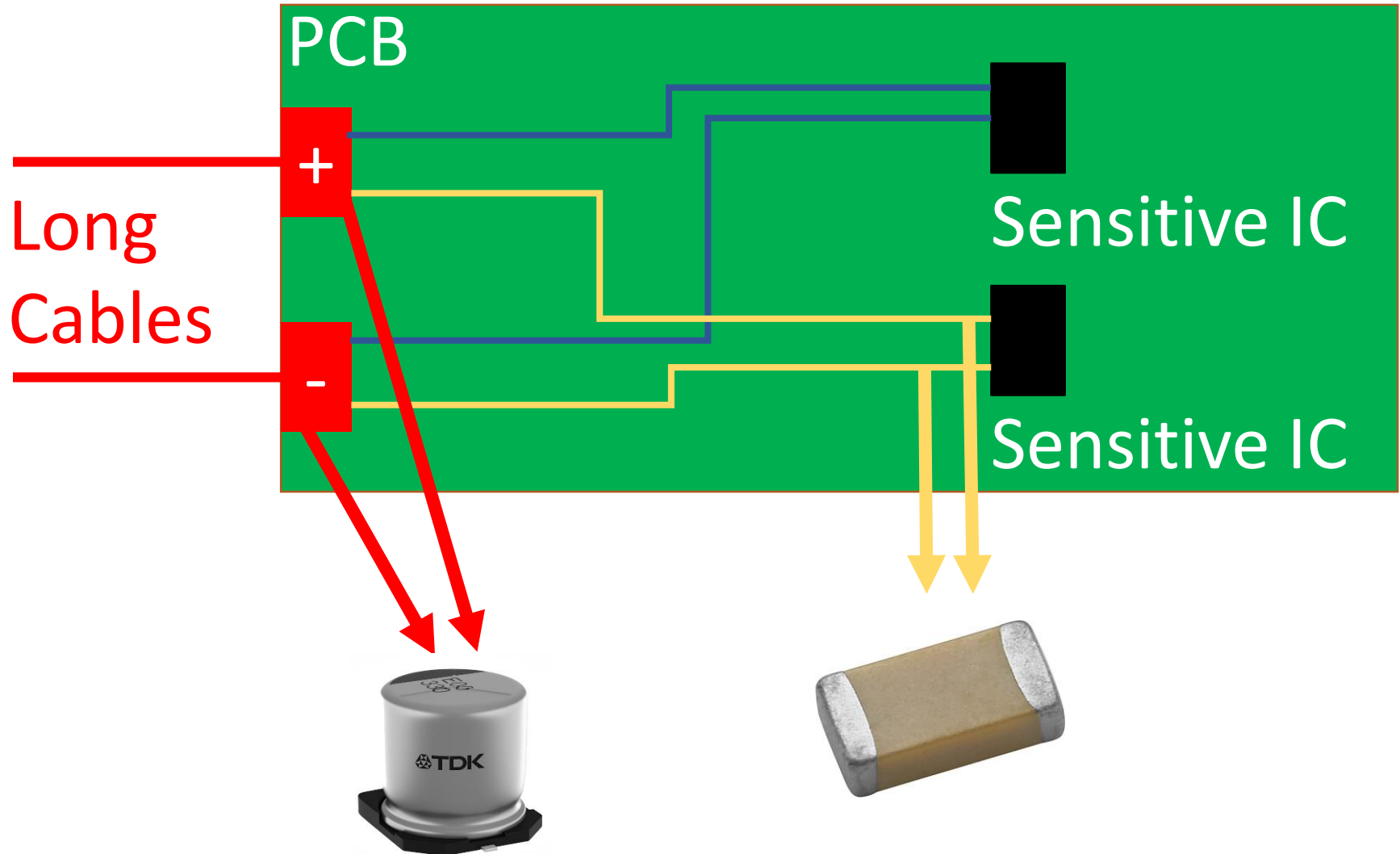


Using Capacitors - Decoupling



Who is going to supply sudden current demands?

Using Capacitors - Decoupling



Provide closest
path for the
highest frequency
current

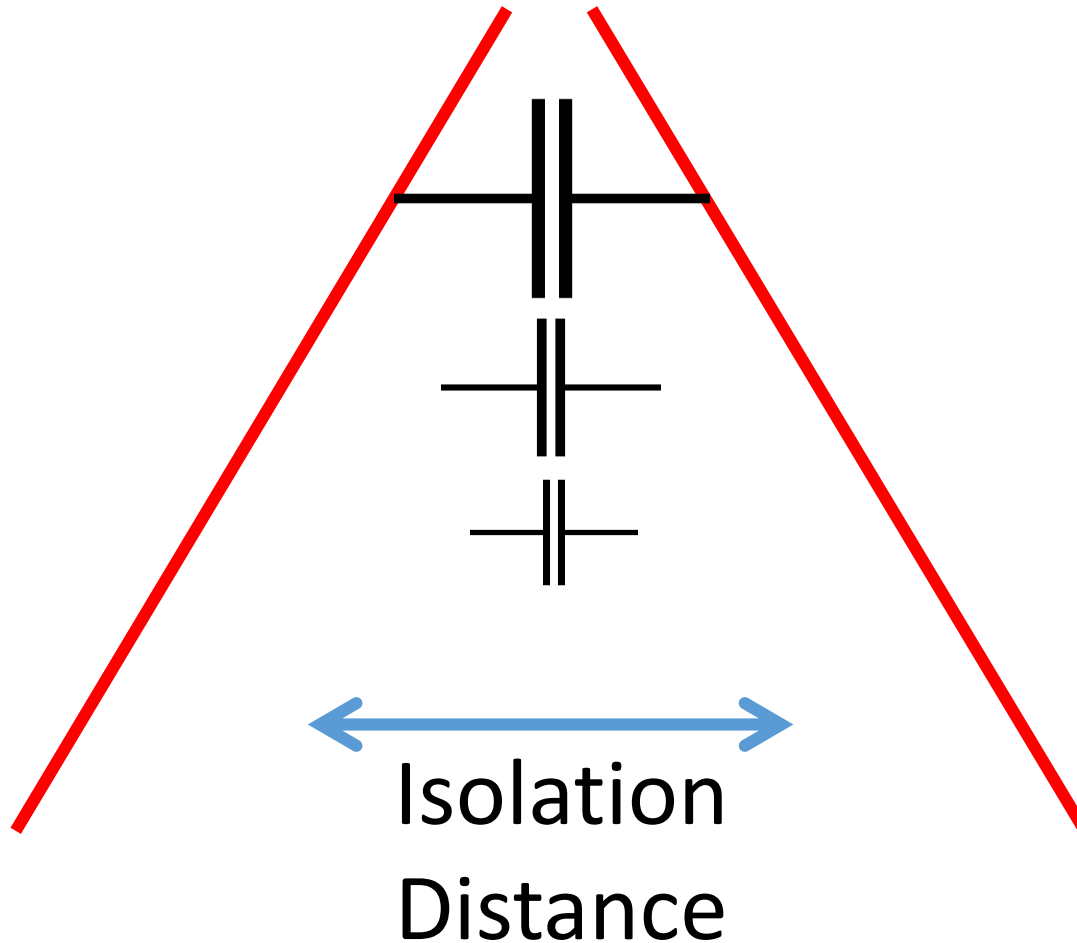
Isolation

We sometimes need to isolate nets from each other to satisfy safety requirements and to improve signal quality.

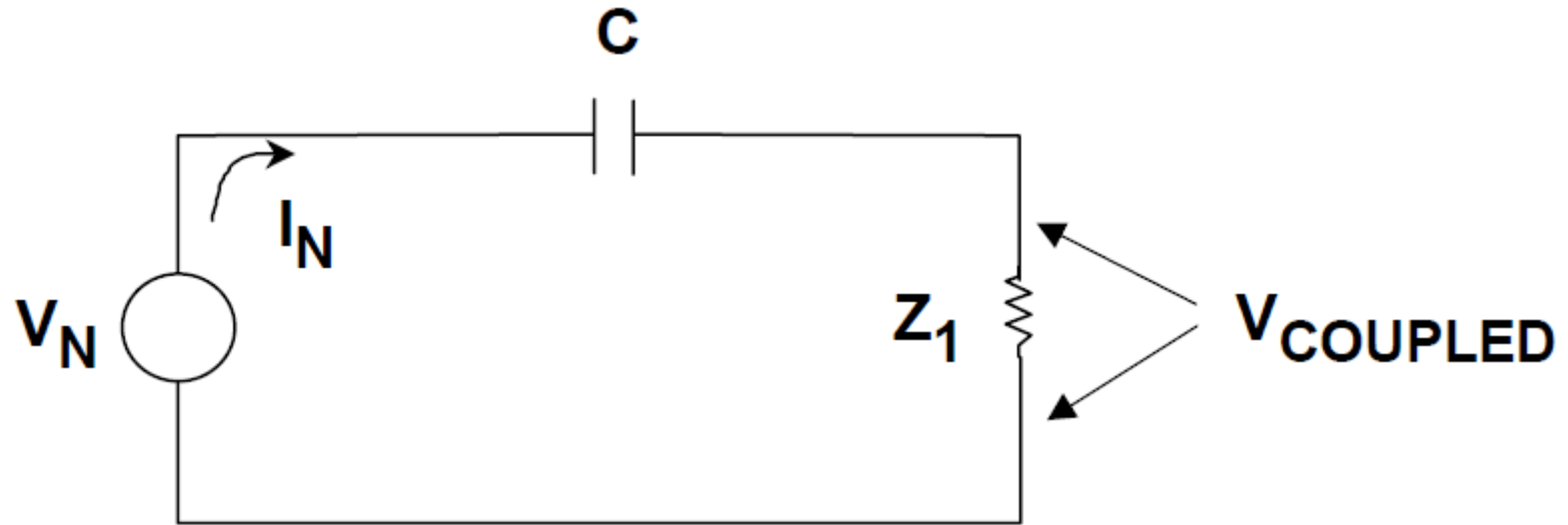
All galvanic isolations have a capacitive equivalent circuitry. Therefore, even though you have separated two nets from each other actually they are connected with a capacitor.

Isolation

$$Capacitance = \frac{k\epsilon_0 A}{d}$$



Isolation

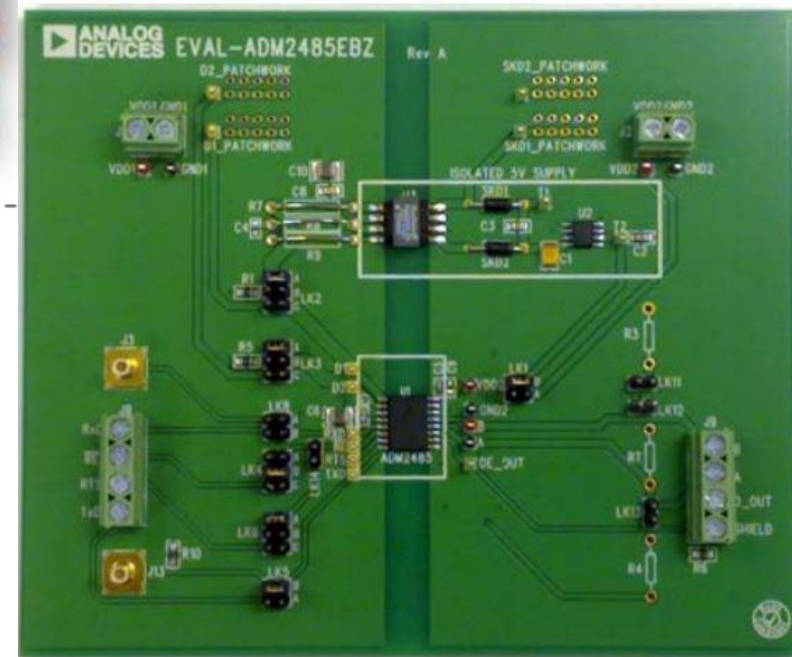
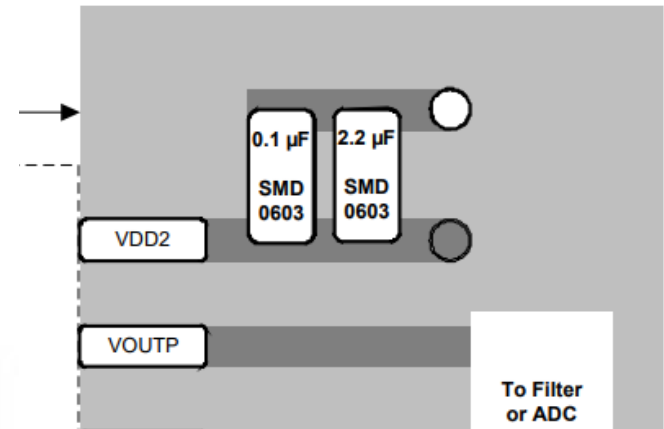


Z_1 = CIRCUIT IMPEDANCE

$Z_2 = 1/j\omega C$

$$V_{\text{COUPLED}} = V_N \left(\frac{Z_1}{Z_1 + Z_2} \right)$$

Isolation



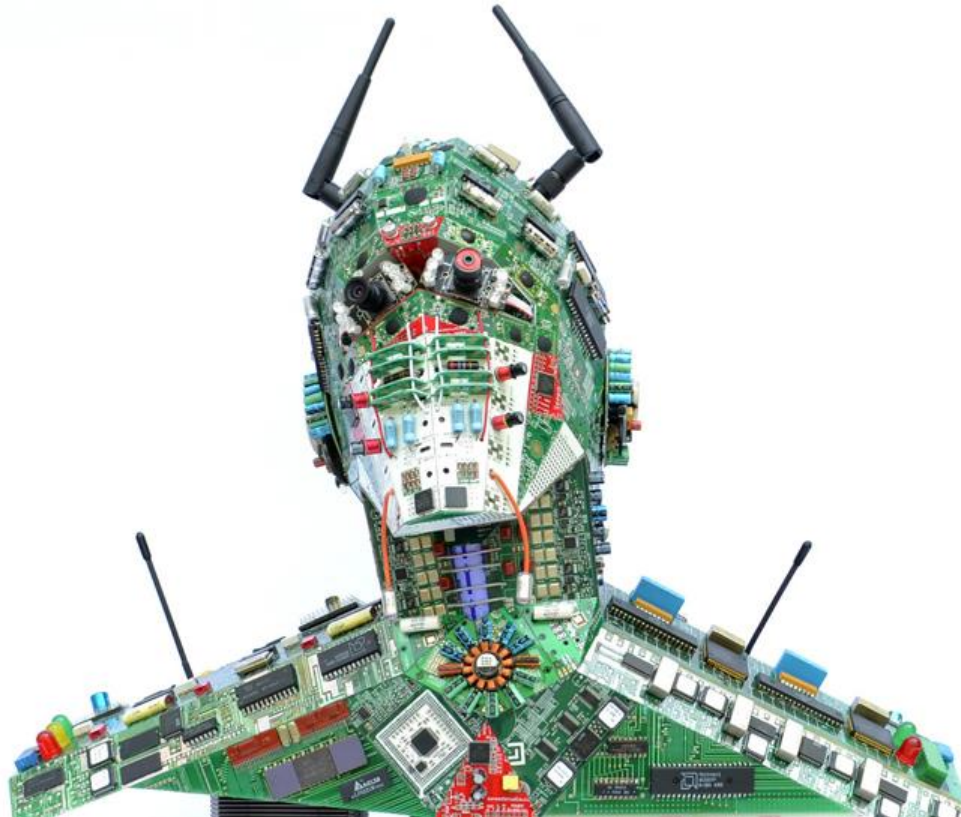
LEGEND

- Copper Pour and Traces
- High-Side Area
- Low-Side Area
- Via to Ground Plane
- Via to Supply Plane

To sum up

- Designing PCB is not just connecting nets to each other
- PCB design should start considering from last to first
- It totally depends on experience. If you are feeling uncomfortable for now, you will be much better soon

Questions?



Do you know that PCBs are recyclable?

References

- <https://www.wellpcb.com/pcb-layers.html>
- <https://en.wikipedia.org/wiki/FR-4>
- https://www.pcbway.com/pcb_prototype/Metal_core_PCBs.html
- <http://www.circuitbasics.com/make-custom-pcb/>
- Clearance and Creepage, David P. Scopelliti, D-Scope-Tech, Inc.
- <https://jlcpcb.com/capabilities/Capabilities>
- <https://www.pcbway.com/pcb-service.html>
- <http://www.ti.com/lit/ds/symlink/amc1301-q1.pdf>